

Optimal Dynamic Selection Under Costly Evaluation: Evidence from Graduate Admissions

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October 8, 2026

Motivation

- **Economic Problem:** Selecting best options under costly evaluation
 - Determining each option's true quality costs decision-maker time and effort
 - Low-cost screening of all options $\Rightarrow$ high-cost evaluation of surviving options
 - **Examples:** Admissions and hiring, clinical trials, matching platforms
- **Why Graduate Admissions:** Benefits graduate programs and future applicants
 - Graduate programs invest time and money into students' development
 - Programs may be undervaluing applicants with certain characteristics
 - Future such applicants may receive currently-withheld opportunities

Research Questions

- ① Are admissions committees optimizing, or are they making mistakes?
 - Goal is to accept applicants with highest success probabilities who will matriculate
 - Per literature, measure success with program completion and job placement ranking
 - Predictors of acceptance only (success only) may be overweighted (underweighted)
 - ② If they are making mistakes, how can they make better admissions decisions?
 - In turn, by changing decision rules, how much better of a cohort can they admit?
 - Look for applicants who were rejected, would have matriculated, and succeeded
- ★ Methods apply to undergraduate/professional admissions, hiring, and other settings

Methods and Results

- Build structural models of admissions committee's decision problem that support:
 - Statistical tests of whether committee's decision rule aligns with desired objective
 - Counterfactual simulations quantifying gain from deviating to model's decision rule
- Estimate model parameters with admissions data from a major research university
 - For most applicants, estimated success probabilities align with committee's decisions
 - Reweighting some applicant characteristics could have increased admitted cohort's average success probability by 4 percentage points without reading more files

Contribution to Literature

- **Graduate Admissions:** Regress acceptance and success on applicant characteristics
 - Krueger & Wu (2000); Athey, Katz, Krueger, Levitt & Poterba (2007); Stock, Finegan & Siegfried (2006–2015); Bai, Esche, MacLeod & Shi (2022)
- **Optimal Dynamic Selection Under Costly Evaluation:**
 - **Sequential Search:** McCall (1970); Mortensen (1970); Weitzman (1979)
 - **Rational Inattention:** Sims (2003); Matějka & McKay (2015); Caplin & Dean (2015)
 - **Optimality:** Cho & Rust (2010); Kleinberg, Lakkaraju, Leskovec, Ludwig & Mullainathan (2018); Rambachan (2024); Anderson, Rosen, Rust & Wong (2025)
- ★ **This Paper:** Builds and estimates dynamic selection models of admissions, which relate acceptance effects to success effects through the committee's problem

Synthetic Data Protocol (IRB-Approved)

① Administrative office receives data from university departments

- Algorithmically extracts variables from textual files, such as recommendation letters

② Synthetic data (Patki et al. 2016) used for model development

- Estimate joint distribution of variables, and simulate synthetic dataset from it
- Rows do not correspond to real subjects, preventing identity disclosure
- Cells with < 5 comparable observations binned to prevent attribute disclosure

③ Administrative office executes final estimation code on actual data

- Only summary statistics and parameter estimates are returned to researcher

★ **Contribution:** Privacy-protecting framework for researchers studying sensitive data

Outline

- 1 Regression Analysis
- 2 Structural Models
- 3 Structural Estimation
- 4 Optimality and Gains

Data Description

- **Sample:** Applicants to research university's economics PhD program from 2015–2019
- **Outcomes:** Measures of admission and success
 - **Admission:** Read File $\Rightarrow$ Accept $\Rightarrow$ Matriculate $\Rightarrow$ Success
 - **Success:** Completion and placement category/quality for **all** applicants, with academia-equivalent rankings for non-academic placements (Krueger & Wu 2000)
- **Covariates:** Objective vs. subjective variables
 - **Objective:** Demographics and performance measures, such as GRE scores
 - **Subjective:** Textual processing of recommendation letters, statements of purpose, CVs, and supplemental forms using dictionaries to construct variables

Regression Results

- **Method:** Logistic regressions for acceptance, matriculation, and success
 - **Success:** *Top 100 Placement* = 1 for 20% of applicants vs. 34% of matriculants
- **Results:** Different independent variables best predict each dependent variable
 - **Accept:** *Committee Score* is best predictor. Other predictors: demographics, *GRE Quantitative Percentile*, *Work Experience*, *CV Coding* (negative)
 - **Matriculate:** Predictors of acceptance often negatively predict matriculation. Best predictor of matriculation is *Open House* attendance
 - **Success:** *PhD Program Rank* is best predictor. Rank held equal, other predictors: *Age*, *TOEFL/IELTS*, *Work Experience*, *Essay Positive Terms (%)*

Acceptance vs. Success AMEs

Dependent Variable	Accept	T100 Placement
Winsorized Age	-0.0001	0.0053*
Female	0.0163**	-0.0032
International Asian	-0.0214***	-0.0127
GRE Quantitative Percentile	0.0011**	0.0006
TOEFL/IELTS	-0.0022	0.0409*
Work Experience	0.0140**	0.0358**
Committee Score	0.0788***	0.0007
CV Coding	-0.0044*	0.0125
Essay Economics Terms (%)	0.0001	-0.0104
Essay Positive Terms (%)	-0.0013	0.0123*

» Insufficiency of Comparing AMEs

» Diversity Objective

» Regressions vs. Model

- Admissions literature regresses acceptance and success on variables in applications, and qualitatively compares the effects
- While such comparisons are of interest, they are insufficient to assess optimality
- Need model of admissions to relate acceptance effects to success effects

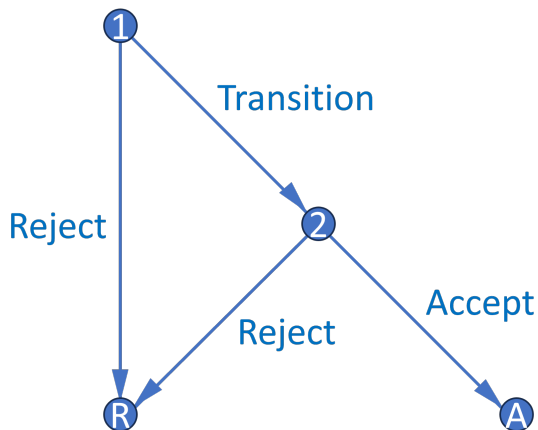
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Why Structural Models?

- **Reason 1:** Test whether committee's decision rule aligns with desired objective
 - Maximize admitted cohort's quality, considering matriculation probabilities
- **Reason 2:** Find counterfactual decision rule in which committee optimizes
 - Quantify gains from deviating from actual to counterfactual decision rule

Dynamics



- **Round 1:** See only objective variables. Transition to Round 2 or reject
- **Round 2:** See survivors' objective and subjective variables. Accept or reject
- Assume for now number of applicants transitioned and accepted is fixed
- Matriculation model will account for success and matriculation probabilities

Portfolio Choice Problem

For each applicant n , choose $d_{1,n} \in \{\text{Transition, Reject}\}$ and $d_{2,n} \in \{\text{Accept, Reject}\}$:

Round 2:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}}, \text{ such that: } \sum_{n=1}^{N_T} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}} \leq N_A,$$

Round 1:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N \underbrace{\frac{EV_{2,n}(x_n)}{\mathbb{E}[\text{Round 2 Value}]}}_{\text{Transition}} \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}}, \text{ such that: } \sum_{n=1}^N \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}} \leq N_T,$$

where x = objective variables, z = subjective variables, $y \in \{\text{Success, Failure}\}$,
 N = number of applicants, (N_T, N_A) = max number of files read/acceptances

Portfolio Choice to Cutoff Rule

- How to estimate model, given so many possible combinations of applicants?
- Assuming no complementarities, portfolio choice problem has cutoff rule solution
- **Round 2:** Accept transitioned applicant if success probability exceeds Round 2 cutoff

$$d_{2,n}^* = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) > \text{Cutoff}_2 \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) = \text{Cutoff}_2 \\ R & \text{if } P(y_n = S | x_n, z_n) < \text{Cutoff}_2 \end{cases}$$

- **Round 1:** Transition applicant if Round 2 expected value exceeds Round 1 cutoff

$$d_{1,n}^* = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \text{Cutoff}_1 \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \text{Cutoff}_1 \\ R & \text{if } EV_{2,n}(x_n) < \text{Cutoff}_1 \end{cases}$$

Cutoff Rule Model

- ★ **Goal:** Use choices and outcomes data to estimate success probabilities, while respecting constraint on number of files committee can read
- ★ **Legend:** Estimated model objects vs. Choices and outcomes data

Cutoff Rule Model, Round 2

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S | t) \mathbb{1}(d_2 = R)}_{\text{If reject, get Round 2 Cutoff}}$$

- **States:** (x, z) = objective and subjective variables, t = year, $\bar{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, (y_1^*, y_2^*) = marginal outcomes

Cutoff Rule Model, Round 1

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2) | x, t, \bar{k}] \mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Round 2 Value}]} + \underbrace{L(y_1^* = S | t) \mathbb{1}(d_1 = R)}_{\text{If reject, get Round 1 Cutoff}}$$

- **States:** (x, z) = objective and subjective variables, t = year, $\bar{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, (y_1^*, y_2^*) = marginal outcomes

Conditional Choice Probabilities

- **Round 2:** Higher payoff from accepting $\Rightarrow$ higher probability of accepting

$$P(d_2 = A|x, z, t, \bar{k}) = \frac{\exp(u_2(\cdot, A)/\sigma)}{\exp(u_2(\cdot, A)/\sigma) + \exp(u_2(\cdot, R)/\sigma)}$$

- **Round 1:** Higher payoff from reading file $\Rightarrow$ higher probability of reading file

$$P(d_1 = T|x, t, \bar{k}) = \frac{\exp(u_1(\cdot, T)/\sigma)}{\exp(u_1(\cdot, T)/\sigma) + \exp(u_1(\cdot, R)/\sigma)}$$

Portfolio Choice with Matriculation

Round 2:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}},$$

such that: $\sum_{n=1}^{N_T} \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}} \leq N_M,$

Round 1:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N \underbrace{EV_{2,n}(x_n)}_{\mathbb{E}[\text{Round 2 Value}]} \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}}, \text{ such that: } \sum_{n=1}^N \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}} \leq N_T,$$

where (x, z) = variables, $y \in \{\text{Success, Failure}\}$, $m \in \{\text{Matriculate, Decline}\}$,
 N = number of applicants, (N_T, N_M) = max number of files read/matriculants

Cutoff Rule with Matriculation

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}},$$

such that: $\sum_{n=1}^{N_T} \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}} \leq N_M,$

Theorem 1

For large N_T , Round 2 problem has a cutoff rule solution on **success** probabilities.

- Likely matriculants more valuable, but also more expensive $\Rightarrow$ effects cancel
- Matriculation probabilities matter in Round 1, especially for low-ranked programs

Cutoff Rule with Matriculation, Round 2

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S|x, z, t, \bar{k})\mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S|t)\mathbb{1}(d_2 = R)}_{\text{If reject, get Round 2 Cutoff}}$$

- **States:** (x, z) = objective and subjective variables, t = year, $\bar{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, $m \in \{\text{Matriculate, Decline}\}$, (y_1^*, y_2^*, m_1^*) = marginal outcomes

Cutoff Rule with Matriculation, Round 1

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2) | x, t, \bar{k}] \mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Round 2 Value}]} + \underbrace{L(y_1^* = S, m_1^* = M | t) \mathbb{1}(d_1 = R)}_{\text{If reject, get Round 1 Cutoff}}$$

- **States:** (x, z) = objective and subjective variables, t = year, $\bar{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, $m \in \{\text{Matriculate, Decline}\}$, (y_1^*, y_2^*, m_1^*) = marginal outcomes

Importance of Matriculation

$$\lim_{\sigma \rightarrow 0} EV_2 = \mathbb{E}_{\tilde{z}}[\max\{P(y = S|x, \tilde{z}, t, \bar{k}) - P(y_2^* = S|t), 0\}P(m = M|x, \tilde{z}, t)|x, t, \bar{k}]$$

- Matriculation matters when expected success probability $\gg$ Round 2 cutoff
- **Upshot:** Low-ranked programs must consider matriculation in Round 1

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Maximum Likelihood Estimation

$$\begin{aligned}
 \mathcal{L}_N^U(\theta) = & \prod_{n=1}^N \underbrace{f(z_n | x_n, t_n; \theta_F)}_{\text{Density of } z|x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}} \\
 & \prod_{n=1}^N \underbrace{P_{d_{1,n}}(\theta_T)^{\mathbb{1}(d_{1,n}=T)} (1 - P_{d_{1,n}}(\theta_T))^{\mathbb{1}(d_{1,n}=R)}}_{\text{Round 1 conditional choice probabilities}} \prod_{n=1}^{N_T} \underbrace{P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Round 2 conditional choice probabilities}}
 \end{aligned}$$

- In unrestricted likelihood $\mathcal{L}^U$, parameterize CCPs with flexible logistic specifications
- In restricted likelihood $\mathcal{L}^R$, use model CCPs, which depend on structural parameters
- $\mathcal{L}^R$ aligns success probabilities implied by outcomes with those implied by CCPs

Cutoff Probability Identification

- Identify each year's cutoffs by how selective committee is that year
 - Fewer acceptances and stronger applicants increase cutoff
 - More acceptances and weaker applicants decrease cutoffs
- While selectivity identifies cutoffs, sorting improves restricted likelihood

Theorem 2

Cutoff probability estimate is such that number of acceptances N_A equals expected N_A . For any cutoff and N_A , likelihood is highest when committee accepts best N_A applicants.

Restricted Estimates

	Round 1 Cutoff	Round 2 Cutoff	Sigma
2015	0.3731 (0.3067, 0.4539)	36.58% (29.81%, 43.92%)	
⋮	⋮	⋮	
2019	0.3887 (0.3268, 0.4624)	38.14% (31.89%, 44.81%)	
Sigma			0.0020 (0.0013)

- Round 1 cutoffs relative to EV_2 . Round 2 cutoffs relative to success probabilities
- Cutoffs stable across years. $\hat{\sigma}$ small $\Leftrightarrow$ distorted restricted success probabilities

Restricted Estimates with Matriculation

	Round 1 Cutoff	Round 2 Cutoff	Sigma
2015	0.0071 (0.0026, 0.0194)	36.35% (29.47%, 43.84%)	
⋮	⋮	⋮	
2019	0.0095 (0.0035, 0.0255)	37.41% (31.04%, 44.26%)	
Sigma			0.0020** (0.0010)

- With matriculation, EV_2 's functional form yields lower Round 1 cutoffs

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Deviation Gains

- Suppose committee adopted model's decision rule with $\sigma = 0$ and N_T unchanged
- Would average success probability of acceptance/matriculation set have increased?
 - 1 Select applicants with model, and determine its acceptance/matriculation set
 - 2 Compute average success probability of its acceptance/matriculation set
 - 3 Compare average success probability to that of set chosen by committee's rule
- Plug $\hat{\theta}_S^U$ into model because $\hat{\theta}_S^R$ may be distorted to rationalize suboptimal behavior

Deviation Gains Test

- Is deviation gain significant? Do z test of $H_0: g(\hat{\theta}_S^U) = 0$ vs. $H_1: g(\hat{\theta}_S^U) > 0$
 - ① Gain is approximately normal, so test statistic is standardized gain. **Or:**
 - ② Compute average success probabilities across B perturbations $\tilde{\theta}_{S,b}^U$ of $\hat{\theta}_S^U$

Theorem 3

P-value is fraction of perturbations with model's $\overline{P(\text{Success})}$ less than committee's.

- Model's set must be exogenously chosen, e.g. train and test on different data
 - If not, gain will be biased upward, and test will be biased toward rejection
 - In bigger dataset, such as hiring at large company, holdout set is feasible

Deviation Gains Correction

- Observed deviation gain is combination of true gain and “winner’s curse”

$$\begin{aligned} g(\hat{\theta}_S^U) &= \underbrace{\bar{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\hat{\theta}_S^U)}_{\text{Model}} - \underbrace{\bar{P}_{\mathcal{A}^U}(\hat{\theta}_S^U)}_{\text{Committee}} \\ &= \underbrace{\bar{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\hat{\theta}_S^U) - \bar{P}_{\mathcal{A}^R(\theta_S^0)}(\hat{\theta}_S^U)}_{\text{Winner's curse}} + \underbrace{\bar{P}_{\mathcal{A}^R(\theta_S^0)}(\hat{\theta}_S^U) - \bar{P}_{\mathcal{A}^U}(\hat{\theta}_S^U)}_{\text{True gain at } \hat{\theta}_S^U} \end{aligned}$$

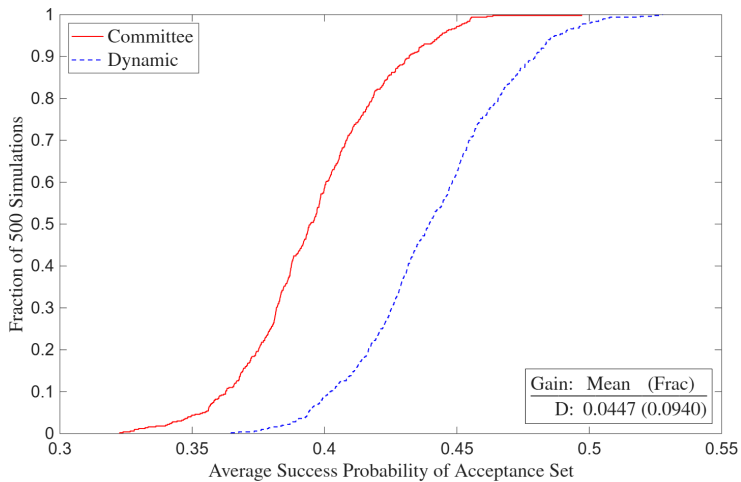
- Subtract estimate of winner’s curse from observed deviation gain

$$\widehat{\text{Winner's curse}} = \frac{1}{B} \sum_{b=1}^B [\bar{P}_{\mathcal{A}^R(\tilde{\theta}_{S,b}^U)}(\tilde{\theta}_{S,b}^U) - \bar{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\tilde{\theta}_{S,b}^U)]$$

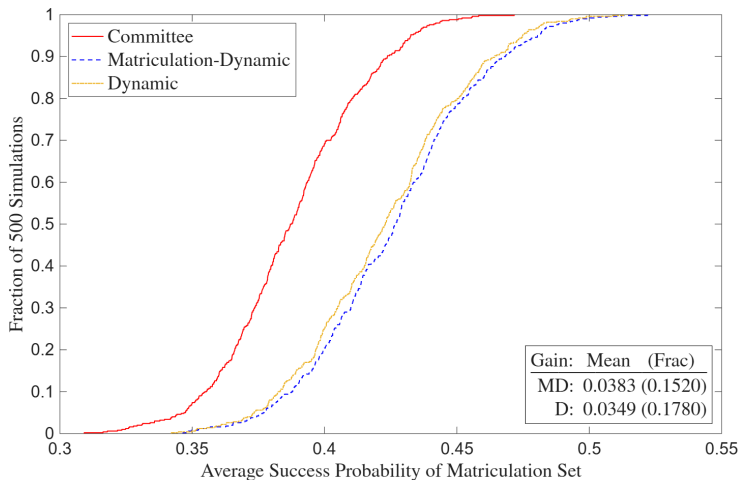
Theorem 4

Correction is exact under certain conditions and removes at least $\sqrt{2} - 1$ of the curse.

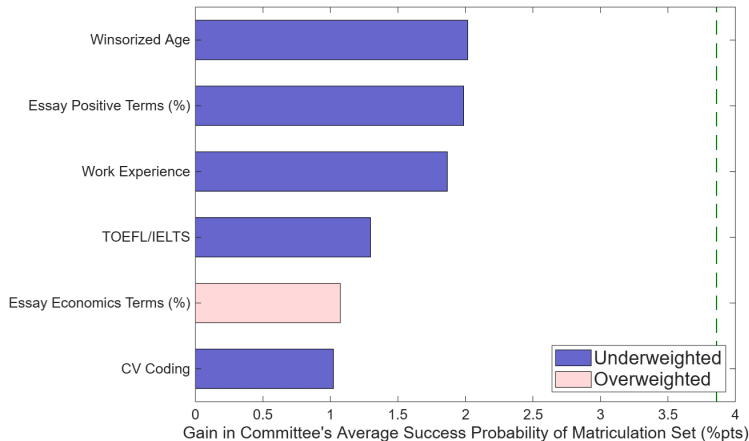
Deviation Gains (Acceptance)



Deviation Gains (Matriculation)

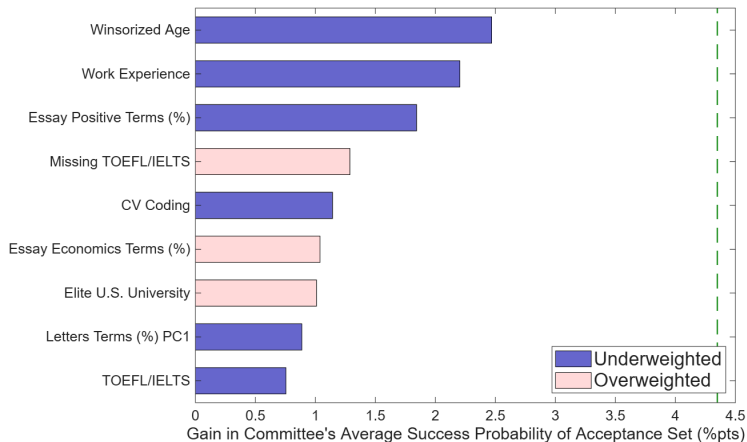


Deviation Gains Attribution (Matriculation)



- 1 Experience
- 2 English Proficiency
- 3 Essay Subtext
- 4 Industry Background

Deviation Gains Attribution (Acceptance)



- 1 Experience
- 2 English Proficiency
- 3 Essay/Letters Subtext
- 4 Industry Background
- 5 Undergraduate

» Full Deviation Gains Attribution (Acceptance)

» Deviation Gains Attribution Table (Acceptance)

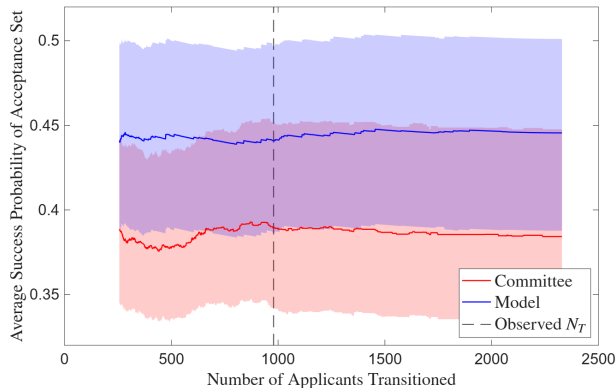
Regressions vs. Model

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Essay Economics Terms (%)	0.0001	-0.0104
Essay Positive Terms (%)	-0.0013	0.0123*

- Blue = underweighted, red = overweighted, white = neither
- AMEs don't reveal if committee deviates from optimal ranking near the cutoff

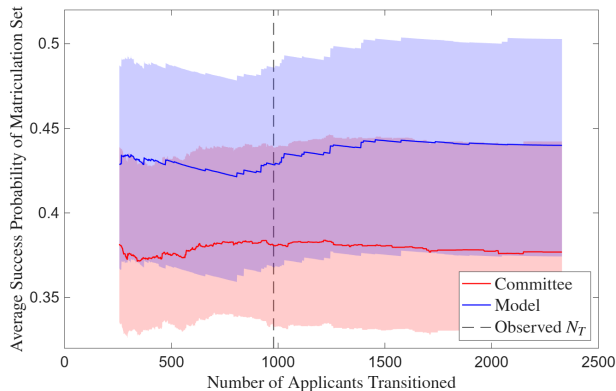
» Acceptance vs. Success AMEs

Benefit of Reading More Files (Acceptance)



- Suppose committee's curve is increasing and concave
- $\max_{n \in [N_A, N]} VN_A \bar{P}(n) - cn$
- **FOC:** $V = c / (N_A \bar{P}'(N_T))$
- V is dollar value of an additional expected success

Benefit of Reading More Files (Matriculation)



- Suppose committee's curve is increasing and concave
- $\max_{n \in [N_A, N]} VN_M \bar{P}(n) - cn$
- **FOC:** $V = c / (N_M \bar{P}'(N_T))$
- V is dollar value of an additional expected success

Conclusion

- **Economic Problem:** Selecting best options under costly evaluation
- **Application:** Graduate admissions using synthetic data protocol to protect privacy
- **Contribution:** Build models to assess optimality and quantify deviation gains
 - ① For most applicants, estimated success probabilities align with committee's decisions
 - ② Reweighting some applicant characteristics could have increased admitted cohort's average success probability by 4 percentage points without reading more files
 - ③ Success is hard to predict, but even modest gains in selection are worthwhile
- ★ Methods apply to undergraduate/professional admissions, hiring, and other settings

Thank You!

A1: Synthetic Rows $\neq$ Actual Applicants

Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
0	0	0	0	0	70	1	0	1	0	.	27
0	0	0	0	0	92	1	1	0	1	.	150
0	0	0	0	0	85	1	1	0	0	.	80
0	0	0	0	0	72	1	0	0	1	.	51
0	0	0	0	0	74	0	1	0	0	.	150
0	0	0	0	0	79	0	0	0	1	.	26
1	1	0	0	1	97	0	0	1	1	8	43
0	0	0	0	0	76	1	0	0	0	.	17
1	1	0	0	0	75	0	0	0	0	3	70
0	0	0	0	0	75	0	0	1	0	.	150



Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
1	1	0	0	0	92	0	0	1	1	8	31
0	0	0	0	0	83	0	0	0	1	.	60
0	0	0	0	1	94	0	1	0	1	.	19
1	1	1	1	0	90	0	0	1	1	5	37
0	0	0	0	1	86	0	0	1	0	.	52
1	0	1	0	1	91	1	1	0	0	2	29
1	1	0	0	0	86	0	0	0	1	5	71
1	1	0	0	0	98	0	1	0	0	5	21
1	1	0	0	1	88	1	1	0	1	4	37
0	0	0	0	0	92	1	1	0	0	.	150

A1: Dependent Variables

Admission	Success
1 (Transition)	1 (Complete Program)
<i>1 (Accept)</i>	<i>1 (Academic Placement)</i>
1 (Waitlist)	1 (Government Placement)
1 (Open House)	1 (Consulting Placement)
<i>1 (Matriculate)</i>	<i>1 (Tech Placement)</i>
1 (Attend Program)	1 (Other Placement)
Program Rank	1 (No Placement)
	Placement Rank
	1 (Top 100 Placement)

- Economics-only variables bold, non-economics-only variables italic
- U.S. News PhD program rankings, IDEAS/RePEc placement rankings
 - For non-tenure track, use weighted average of university's and maximum rankings
 - For non-academia, use average of ChatGPT, Claude, and Gemini's [rankings](#)

A1: Objective Variables

Demographic	Performance
Year	GRE Verbal Percentile
1 (<i>Covid Year</i>)	GRE Quantitative Percentile
Winsorized Age	GRE Analytic Percentile
1 (Female)	Undergraduate GPA
1 (U.S.)	TOEFL/IELTS
1 (<i>U.S. African/Hispanic/Native</i>)	1 (Elite U.S. University)
1 (<i>U.S. Asian</i>)	1 (Elite U.S. Liberal Arts College)
1 (<i>U.S. Other</i>)	1 (Other U.S. University/College)
1 (International Asian)	1 (Elite Foreign University)
1 (International Other)	1 (Other Foreign University)
	1 (Graduate Degree)
	1 (Work Experience)
	#Professor Recommenders
	#Prolific Recommenders
	<i>Concentration Dummy Variables</i>

A1: Subjective Variables

Score/Recommendation Letters	Supplement/CV/Essay
Committee Score	#Math Courses
Letters Length	1 (Advanced Math)
Positive Terms (%) (YS 2012)	CV Length
Negative Terms (%) (YS 2012)	CV Coding
Standout Terms (%) (BEMS 2022)	CV 1 (Publication/Working Paper)
Ability Terms (%) (BEMS 2022)	Essay Length
Research Terms (%) (BEMS 2022)	Essay 1 (Specific Professor)
Grindstone Terms (%) (BEMS 2022)	Essay Economics Terms (%)
Teaching Terms (%) (BEMS 2022)	Essay Research Terms (%)
Communal Terms (%) (BEMS 2022)	Essay Positive Terms (%)
Agentic Terms (%) (MHM 2009)	Essay Negative Terms (%)
	Field Dummy Variables

- Dictionaries from Madera, Hebl & Martin (2009), Young & Soroka (2012), and Bai, Esche, MacLeod & Shi (2022)
- *#Math Courses* and *Advanced Math* objective after 2020

A1: Academia-Equivalent Rankings (Government)

- **European Institutions:** European Central Bank (40), HM Treasury (80)
- **Foreign Central Banks:** Banco Central do Brasil (93), Banco de México (93), Bank of Canada (80), Bank of England (67), Bank of Israel (93), Bank of Italy (80), Bank of Japan (80), Bank of Spain (80), Banque de France (80), Central Bank of Chile (93), Central Bank of Colombia (93), Central Bank of Korea (93), Central Bank of Turkey (93), Norges Bank (80), Reserve Bank of Australia (80), Reserve Bank of India (93), Reserve Bank of New Zealand (93), Sveriges Riksbank (80), Swiss National Bank (80)
- **Multilaterals/International:** African Development Bank (93), Asian Development Bank (93), Bank for International Settlements (53), European Bank for Reconstruction and Development (93), European Investment Bank (93), Inter-American Development Bank (93), International Monetary Fund (40), Statistics Canada (107), World Bank (40)
- **Policy Research Institutes:** American Enterprise Institute (93), American Institutes for Research (107), Becker Friedman Institute (80), Brookings Institution (80), Center for Global Development (93), Center for Migration Studies in New York (112), Center for Strategic and International Studies (107), Cowles Foundation (80), Harris School Policy Labs (93), Hoover Institution (80), IDinsight (112), Innovations for Poverty Action (93), J-PAL (80), MDRC (107), National Bureau of Economic Research (67), Peterson Institute for International Economics (80), Pew Research Center (107), RAND Corporation (80), Resources for the Future (93), Urban Institute (107)
- **U.S. Executive/Cabinet:** Department of Commerce (107), Department of Justice (80), Department of Labor (107), Department of the Treasury (80)
- **U.S. Executive Policy Offices:** Council of Economic Advisers (80), Office of Management and Budget (93)
- **U.S. Federal Reserve System:** Board of Governors (40), Atlanta (80), Boston (80), Chicago (80), Cleveland (80), Dallas (80), Kansas City (80), Minneapolis (80), New York (67), Philadelphia (80), Richmond (80), San Francisco (80), St. Louis (80)
- **U.S. GSE/Housing Finance:** Fannie Mae (120), Freddie Mac (120)
- **U.S. Independent Agencies:** Commodity Futures Trading Commission (93), Federal Deposit Insurance Corporation (93), Federal Energy Regulatory Commission (93), Federal Housing Finance Agency (93), Federal Trade Commission (80), Office of Financial Research (93), Securities and Exchange Commission (93)
- **U.S. Legislative Committees:** Congressional Budget Office (80), Congressional Research Service (107), Joint Committee on Taxation (93), Joint Economic Committee (107)
- **U.S. Science and Research:** National Science Foundation (107)
- **U.S. Statistical Agencies:** Bureau of Economic Analysis (93), Bureau of Labor Statistics (93), Census Bureau (93)

A1: Academia-Equivalent Rankings (Consulting/Tech/Other)

- **Consulting:** Aecon Consulting (130), AlixPartners (125), Analysis Group (120), Bain & Compnay (120), Bates White (120), Berkeley Research Group (120), Boston Consulting Group (120), Brattle Group (120), Charles River Associates (120), Coleman Research (130), Compass Lexecon (120), Cornerstone Research (120), Deloitte Economic and Financial Advisory (125), Econic Partners (130), Economists Incorporated (120), Edgeworth Economics (120), Ernst & Young Economic Advisory (125), Frontier Economics (120), FTI Consulting (120), Guidehouse (130), KPMG Economics & Valuation (125), Matrix Economics (130), McKinsey & Company (120), Mercer (135), NERA Economic Consulting (120), Oliver Wyman (120), PwC Economics (125), Roland Berger (125), Willis Towers Watson (135)
- **Tech (Athey & Luca 2019):** Airbnb (107), Alibaba (120), Amazon (93), AppNexus (135), Apple (107), Block/Square (120), ByteDance/TikTok (120), CoreLogic (135), Coursera (120), DStillery (135), Didichuxing (125), Digonex (135), DoorDash (107), eBay (107), ECONorthwest (125), Expedia (120), Forkcast (135), Glassdoor (120), Google (93), Granular (135), Groupon (125), Houzz (130), Huawei (120), IBM (120), Indeed (120), ING (130), Instacart (120), Intel (120), Kensho (125), Lending Club (135), LinkedIn (107), Lyft (120), Meta/Facebook (93), Microsoft (93), Netflix (93), Nuna (135), Nvidia (120), Pandora (135), PayPal (120), Pinterest (120), Prattle (135), Quantco (130), Quora (130), Redfin (120), Ripple (135), Roblox (120), Rover (135), Salesforce (120), Shopify (120), Spotify (120), Stripe (107), Trulia (125), Uber (107), Upwork (135), Visa (125), Walmart (125), Wealthfront (135), Yahoo (125), Yelp (125), Zillow (120)
- **Tech Labs:** Amazon Science (40), Apple Machine Learning Research (80), DeepMind (80), Google Research (40), Meta Research (80), Microsoft Research (40), Netflix Research (93), OpenAI Research (80), Uber Research (93)
- **Other:** AIG (135), Bank of America (125), Bloomberg (125), Boeing (135) Capital One (125), Exelon (135), ExxonMobil (135), Ford (135), General Electric (135), General Motors (135), Goldman Sachs (125), JPMorgan Chase (125), Liberty Mutual (135), Moody's Analytics (125), Morningstar (125), Nielsen (135), PNC (135), Progressive (135), S&P Global (125), Shell (135), Swiss Re (130), Toyota (130), Other (135), No Placement (150)

A1: Dictionaries

● Recommendation Letters:

- **Positive/Negative Words:** Young & Soroka (2012)'s Lexicoder Sentiment Dictionary
- **Standout, Ability, Research, Grindstone, Communal Words:** Bai, Esche, MacLeod & Shi (2022)
- **Agentic Words:** assertive, confident, aggressive, ambitious, dominant, forceful, independent, daring, outspoken, intellectual (Madera, Hebl & Martin 2009); **AND** audacious, bold, command, compete, decisive, determine, drive, enterprise, fearless, influential, leader, pioneer, powerful, proactive, resilient, resourceful, self-assured, strategic, tenacious, vision
- **Teaching Words:** academic, assess, clarity, coach, collaborate, communicate, cultivate, curriculum, demonstrate, develop, educate, encourage, engage, enlighten, evaluate, explain, facilitate, foster, guide, impart, inspire, insight, instruct, knowledge, learn, lesson, mentor, nurture, participate, pedagogy, present, scholar, support, teach, train, tutor, workshop

● CVs and Essays:

- **Coding:** c/c++, eviews, fortran, java, jax, julia, latex, matlab, python, r, sas, stata, webscrape
- **Publication:** Journal names from IDEAS/RePEc Aggregate Rankings for Journals
- **Working Paper:** arxiv, center for economic policy research, cepr, econpapers, national bureau of economic research, nber, research papers in economics, repec, social science research network, ssrn, working paper
- **Economics:** applied, asset, behavior, climate, computation, data, development, dynamic programming, economics, education, empirical, environment, experiment, finance, fiscal, game theory, health, immigration, industrial, inequality, international, io, labor, linear programming, macro, math, metrics, micro, monetary, policy, political, poverty, pricing, probability, public, sports, statistic, theory, trade, urban

A1: Inference vs. Prediction

- Challenging to obtain consistent estimates due to omitted variables
 - **Example:** Don't observe effort (e.g. hours worked). Can bias coefficients upward if effort correlated with variables and with success irrespective of those variables
 - Existing literature also suffers from this problem, though BEMS (2022) parse recommendation letters for “grindstone terms” (e.g. depend*, diligen*)
- Research questions emphasize prediction over inference
 - Mullainathan & Spiess (2017) distinguish $\hat{\beta}$ vs. $\hat{y}$ problems
 - If GRE predicts success, committee should use it regardless of why

A1: Sample Selection

- Determinants of success conditional on applying vs. matriculating
- BEMS (2022) prove that if $P(\text{Success}|\text{Skill})$ is an S-curve, then $\hat{\beta}$ will be smaller for more selected samples. Consistent with empirical results
- If there aren't some types of outcomes data (e.g. grades, passing comps) for all applicants, or such data are too hard to obtain, use Heckman correction
 - Let $y_n^* = x_n\beta + u_n$, $d_n = \mathbb{1}(z_n\gamma + v_n > 0)$, and $y_n = d_n y_n^*$
 - **Step 1:** Run probit of d on z and compute inverse Mills $\hat{\lambda}_n = \frac{\phi(z_n\hat{\gamma})}{\Phi(z_n\hat{\gamma})}$
 - **Step 2:** Run OLS or heckprobit of y on $(x, \hat{\lambda})$ to get $\hat{\beta}$
 - **Instruments:** Variables that predict matriculation but not the outcome

A1: Acceptance, Matriculation, and Success AMEs

Dependent Variable	Accept	Matriculate	T100 Placement	T100 Placement
Winsorized Age	-0.0001	-0.0223**	0.0037	0.0053*
Female	0.0163**	-0.0490	-0.0157	-0.0032
U.S.	-0.0013	0.1472*	0.0213	0.0192
International Asian	-0.0214***	0.1644**	0.0298	-0.0127
GRE Quantitative Percentile	0.0011**	-0.0147**	0.0028**	0.0006
Undergraduate GPA	0.0195	-0.1026	-0.0721	-0.1096**
TOEFL/IELTS	-0.0022	-0.0631	0.0219	0.0409*
Work Experience	0.0140**	0.1591**	0.0271	0.0358**
#Prolific Recommenders	0.0039	-0.0426	0.0137*	0.0076
Committee Score	0.0788***	-0.0371	0.0144**	0.0007
Advanced Math	0.0246	-0.1950**	-0.0317	-0.0287
Letters Terms (%) PC1	0.0055	0.0671**	0.0038	0.0159
CV Coding	-0.0044*	-0.0414**	0.0108	0.0125
Essay Economics Terms (%)	0.0001	0.0366**	-0.0175**	-0.0104
Essay Positive Terms (%)	-0.0013	-0.0072	0.0118*	0.0123*
Missing Committee Score	-0.0096		-0.0528**	-0.0277
Open House		0.2615***		
Program Rank				-0.0011***
Missing Program Rank				-0.3965***

Year Dummies

GRE Verbal Percentile

GRE Analytic Percentile

Elite U.S. University

Elite U.S. Liberal Arts College

Elite Foreign University

Other Foreign University

Graduate Degree

#Professor Recommenders

#Math Courses

Letters Length/100

Letters Terms (%) PC2

CV Length/100

CV Publication/Working Paper

Essay Length/100

Essay Specific Professor

Essay Research Terms (%)

Essay Negative Terms (%)

Missing Dummies

A1: Insufficiency of Comparing AMEs

- Cannot assess optimality by comparing AMEs across regressions. Suppose:
 - Work experience has large effect on success, graduate degree smaller effect
 - 25% of applicants have both, 25% experience only, 25% degree only, 25% neither
 - Committee can accept half of applicants, and other half attend different university
- Optimal committee accepts half with experience and ignores degree
 - **Acceptance Regression:** $AME_{\text{Experience}} = 1, AME_{\text{Degree}} = 0$
 - **Success Regression:** $1 > AME_{\text{Experience}} > AME_{\text{Degree}} > 0$
 - Despite optimality, acceptance and success AMEs not equal across regressions
- **Upshot:** Relationship between AMEs must be interpreted through a model

A1: Diversity Objective

- **Demographics:** Squares vs. circles

- **Quality:** High vs. low

High Quality:



Low Quality:



A1: Diversity Objective

- No preference for squares or circles

High Quality:



Low Quality:



» Acceptance vs. Success AMEs

A1: Diversity Objective

- Small preference for squares $\Rightarrow$ consistent with maximizing quality

High Quality:



Low Quality:



►► Acceptance vs. Success AMEs

A1: Diversity Objective

- Large preference for squares $\Rightarrow$ inconsistent with maximizing quality

High Quality:



Low Quality:



►► Acceptance vs. Success AMEs

A1: Wald Tests

- Test if acceptance and/or success AMEs are same across PhD programs
 - **Example:** Does elite undergraduate university predict acceptance and/or success more or less for economics program than for other programs?
 - **Sample:** Applicants to university's economics, government, history, linguistics, and psychology PhD programs from 2020–2023 (when multi-program data is available)
- Can I test if acceptance AMEs equal success AMEs within programs?
 - Predictors of acceptance only (success only) may be overweighted (underweighted), provided committee wants to select based on a “success” model
 - **Problem:** Not a test if admissions decisions are optimal
 - **Solution:** Need model of admissions to assess optimality

A1: Logistic Wald Test Across Programs

Assume $(Y^E \perp\!\!\!\perp Y^{NE})|X$. Let $P(Y_n^E = 1|x_n^E; \theta^E) = \frac{\exp(x_{n,0}^E \theta_0^E + x_{n,1}^E \theta_1^E)}{1 + \exp(x_{n,0}^E \theta_0^E + x_{n,1}^E \theta_1^E)}$,

and $P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE}) = \frac{\exp(x_{n,0}^{NE} \theta_0^{NE} + x_{n,1}^{NE} \theta_1^{NE})}{1 + \exp(x_{n,0}^{NE} \theta_0^{NE} + x_{n,1}^{NE} \theta_1^{NE})}$. Finally, let $N = N_E + N_{NE}$. Then:

$$\mathcal{L}_N^U(y|x; \theta) = \prod_{n=1}^{N_E} P(Y_n^E = 1|x_n^E; \theta^E)^{Y_n^E} [1 - P(Y_n^E = 1|x_n^E; \theta^E)]^{1-Y_n^E} \prod_{n=1}^{N_{NE}} P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE})^{Y_n^{NE}} [1 - P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE})]^{1-Y_n^{NE}}.$$

- $Y \in \{\mathbb{1}(\text{Accept}), \mathbb{1}(\text{Attend Program}), \mathbb{1}(\text{Complete Program}), \mathbb{1}(\text{Good Placement})\}$
- **Test:** Let $g(\hat{\theta}) = \frac{1}{N_E} \sum_{n=1}^{N_E} \hat{P}_n^E (1 - \hat{P}_n^E) \hat{\theta}_1^E - \frac{1}{N_{NE}} \sum_{n=1}^{N_{NE}} \hat{P}_n^{NE} (1 - \hat{P}_n^{NE}) \hat{\theta}_1^{NE}$. Then:
 $W = g(\hat{\theta})' [\nabla_{\theta} g(\hat{\theta}) \hat{V}(\hat{\theta}) \nabla_{\theta} g(\hat{\theta})']^{-1} g(\hat{\theta}) \sim \chi_Q^2$. Compare AMEs per Mood (2010)

A1: Linear Wald Test Across Programs

Assume $(Y^E \perp\!\!\!\perp Y^{NE})|X$, i.e. (Economics $\perp\!\!\!\perp$ Non-Economics) | Predictors.

Let $f(y_n^E|x_n^E; \beta^E, \sigma_E^2) = \frac{1}{\sqrt{2\pi\sigma_E^2}} \exp\left[\frac{-1}{2\sigma_E^2}(y_n^E - x_{n,0}^E\beta_0^E - x_{n,1}^E\beta_1^E)^2\right]$, and

$f(y_n^{NE}|x_n^{NE}; \beta^{NE}, \sigma_{NE}^2) = \frac{1}{\sqrt{2\pi\sigma_{NE}^2}} \exp\left[\frac{-1}{2\sigma_{NE}^2}(y_n^{NE} - x_{n,0}^{NE}\beta_0^{NE} - x_{n,1}^{NE}\beta_1^{NE})^2\right]$. Then:

$$\mathcal{L}_N^U(y|x; \beta, \sigma^2) = \prod_{n=1}^{N_E} f(y_n^E|x_n^E; \beta^E, \sigma_E^2) \prod_{n=1}^{N_{NE}} f(y_n^{NE}|x_n^{NE}; \beta^{NE}, \sigma_{NE}^2).$$

- $Y \in \{\text{Standardized Committee Score, Standardized Placement Rank}\}$
- **Test:** Let $g(\hat{\beta}_1) = \hat{\beta}_1^E - \hat{\beta}_1^{NE}$. (Can also do likelihood-ratio test.)
Then $W = g(\hat{\beta}_1)'[\nabla_{\beta, \sigma^2} g(\hat{\beta}_1) \hat{V}(\hat{\beta}, \hat{\sigma}^2) \nabla_{\beta, \sigma^2} g(\hat{\beta}_1)']^{-1} g(\hat{\beta}_1) \sim \chi_Q^2$
- **Assumption:** Homoskedasticity (i.e. $\forall n, \sigma_n^2 = \sigma^2$)

A1: Additional Questions

- Are there different determinants of different types of success? (AKKLP 2007)
 - Need different skills at different stages in the program, such as grades vs. placement
 - For some programs, initial placement matters less than for economics
- Different determinants for international applicants? (KW 2000)
 - **Admission:** May have different preparation and/or interests
 - **Completion:** Depending on country, may have fewer outside options
 - **Placement:** May return to native country after graduation
- Different determinants over time? (2015 to 2023)
 - COVID-19 and other macro factors; changes in professions (e.g. more empirics)
 - For economics, increased importance of elite predocs and Fed RAships

A2: Models of Admissions

- ① **Dynamic Model:** Committee ranks applicants based on conditional expectation in Round 1 about those who survive to Round 2, reads files of those above Cutoff 1, and rejects the rest. Favors those in Round 1 with objective variables positively correlated with subjective variables that predict success. Then re-ranks survivors based on all variables, accepts survivors above Cutoff 2, and rejects the rest
 - ① **Static Model:** In Round 1, rank applicants based only on objective variables
 - ② **Myopic Model:** Like dynamic, but distribution of z not conditional on x
- ② **Matriculation-Dynamic Model:** In Round 1, rank applicants based on combination of success and matriculation probabilities (largely success probabilities)

A2: Description of Models

- **Round 2:** If accept applicant, get success probability $P(y = S|x, z, t, \bar{k})$, where $\text{Success} = \mathbb{1}(\text{Complete Program})$ or $\mathbb{1}(\text{Good Placement})$. If reject applicant, get $P(y_2^* = S|t) = P(\text{Success})$ of year t 's marginal accepted applicant
- **Round 1:** If reject applicant, get $L/P(y_1^* = S|t) = L/P(\text{Success})$ of year t 's marginal transitioned applicant. Can be less or greater than $P(y_2^* = S|t)$ depending on value placed on committee's time vs. admitting a successful cohort
 - **Dynamic:** If transition applicant, get $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] =$ expected value of Round 2, where z is integrated out conditional on (x, t) based on $F(z|x, t)$
 - **Static:** If transition applicant, get $P(y = S|x, t, \bar{k})$, or $P(\text{Success})$ unconditional on z . It may differ from $P(y = S|x, z, t, \bar{k})$, though it still accounts for z 's effect on y
 - **Myopic:** In $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$, replace $F(z|x, t)$ with $F(z|t)$

A2: Lagrangian (BEMS 2022)

Portfolio Choice Problem:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N P(y_n = S | x_n, z_n) \mathbb{1}(d_n = A), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_n = A) \leq N_A$$

$$\mathcal{L}(d_1, \dots, d_N; \lambda) = \sum_{n=1}^N [P(y_n = S | x_n, z_n) \mathbb{1}(d_n = A) + \lambda \mathbb{1}(d_n = R)] + \lambda(N_A - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(x_n, z_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) = \lambda^* \\ R & \text{if } P(y_n = S | x_n, z_n) < \lambda^* \end{cases}$$

Upshot: λ^* is cutoff probability. Denote it hereafter as $P(y^* = S)$

- $P(y^* = S) \in [P(y_{N_A+1} = S | x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S | x_{N_A}, z_{N_A})]$

A2: Lagrangian with Shocks

Portfolio Choice Problem:

$$\begin{aligned} \max_{(d_1, \dots, d_N) \in \{A, R\}^N} & \sum_{n=1}^N \{ [P(y_n = S | x_n, z_n) + \epsilon_n(A)] \mathbb{1}(d_n = A) + \epsilon_n(R) \mathbb{1}(d_n = R) \}, \\ \text{such that: } & \sum_{n=1}^N \mathbb{1}(d_n = A) \leq N_A. \quad \text{Then: } \mathcal{L}(d_1, \dots, d_N; \lambda) = \\ & \sum_{n=1}^N \{ [P(y_n = S | x_n, z_n) + \epsilon_n(A)] \mathbb{1}(d_n = A) + [\lambda + \epsilon_n(R)] \mathbb{1}(d_n = R) \} + \lambda(N_A - N) \end{aligned}$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(x_n, z_n, \epsilon_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) > \lambda^* + \epsilon_n(R) \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) = \lambda^* + \epsilon_n(R) \\ R & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) < \lambda^* + \epsilon_n(R) \end{cases}$$

Theorem 5

For large N , there is a cutoff rule solution with logistic CCPs. ► Portfolio Choice to Cutoff Rule

A2: Lagrangian, Round 2

Portfolio Choice Problem:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S | x_n, z_n) \mathbb{1}(d_{2,n} = A), \text{ such that: } \sum_{n=1}^{N_T} \mathbb{1}(d_{2,n} = A) \leq N_A$$

$$\mathcal{L}(d_{2,1}, \dots, d_{2,N_T}; \lambda_2) = \sum_{n=1}^{N_T} [P(y_n = S | x_n, z_n) \mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)] + \lambda_2 (N_A - N_T)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N_T\}, d_{2,n}(x_n, z_n | \lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) = \lambda_2^* \\ R & \text{if } P(y_n = S | x_n, z_n) < \lambda_2^* \end{cases}$$

Upshot: λ_2^* is cutoff probability. Denote it hereafter as $P(y_2^* = S)$

- $P(y_2^* = S) \in [P(y_{N_A+1} = S | x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S | x_{N_A}, z_{N_A})]$

A2: Lagrangian, Round 1 (Dynamic)

Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N EV_{2,n}(x_n; \bar{\lambda}_2^*) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n) = \int_{\tilde{z}_n} \max\{P(y_n = S | x_n, \tilde{z}_n), \bar{\lambda}_2^*\} dF_{z|x}(\tilde{z}_n | x_n)$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^* \end{cases}$$

Upshot: λ_1^* is cutoff level. Denote it hereafter as $L(y_1^* = S) \in [EV_{2,N_T+1}, EV_{2,N_T}]$

A2: Lagrangian, Round 1 (Static)

Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N P(y_n = S | x_n) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [P(y_n = S | x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } P(y_n = S | x_n) > \lambda_1^* \\ \{T, R\} & \text{if } P(y_n = S | x_n) = \lambda_1^* \\ R & \text{if } P(y_n = S | x_n) < \lambda_1^* \end{cases}$$

Upshot: λ_1^* is cutoff probability. Denote it hereafter as $P(y_1^* = S)$

- $P(y_1^* = S) \in [P(y_{N_T+1} = S | x_{N_T+1}), P(y_{N_T} = S | x_{N_T})]$

A2: Dynamic Parameterization 1

Unrestricted CCPs:

$$P_{d_1} = P(d_1 = T | x, t; \theta_T) = \frac{\exp(f_T(x, t)\theta_T)}{1 + \exp(f_T(x, t)\theta_T)}$$

$$P_{d_2} = P(d_2 = A | x, z, t; \theta_A) = \frac{\exp(f_A(x, z, t)\theta_A)}{1 + \exp(f_A(x, z, t)\theta_A)}$$

Success Probability:

$$P_y = P(y = S | x, z, t, k; \theta_S) = \frac{\exp(f_S(x, z, t, k)\theta_S)}{1 + \exp(f_S(x, z, t, k)\theta_S)}$$

Cutoff Level/Probability:

$$L_{y_1^*} = L(y_1^* = S | t; \theta_{S_1^*}) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_2^*} = P(y_2^* = S | t; \theta_{S_2^*}) = \frac{\exp(f_{S_2^*}(t)\theta_{S_2^*})}{1 + \exp(f_{S_2^*}(t)\theta_{S_2^*})}$$

A2: Dynamic Parameterization 2

$$\begin{aligned}\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] &= \int_{\tilde{z}} \sigma \log \left(\sum_{\tilde{d}_2 \in \{A, R\}} e^{u_2(x, \tilde{z}, t, \bar{k}, \tilde{d}_2)/\sigma} \right) dF(\tilde{z}|x, t) \\ &= \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_J} \left[\cdot \right] dF_1(\tilde{z}_1|\tilde{z}_2, \dots, \tilde{z}_J, x, t) \cdots dF_J(\tilde{z}_J|x, t)\end{aligned}$$

- **Continuous:** $f_{z|x,t}(z_j|x, t; \beta_{F_{z|x,t,j}}, \sigma_{F_{z|x,t,j}}^2) = \frac{1}{\sqrt{2\pi\sigma_{F_{z|x,t,j}}^2}} \exp \left[\frac{-1}{2\sigma_{F_{z|x,t,j}}^2} (z_j - g(x, t)\beta_{F_{z|x,t,j}})^2 \right]$
- **Binary:** $P(z_j = 1|x, t; \theta_{F_{z|x,t,j}}) = \frac{\exp(g(x, t)\theta_{F_{z|x,t,j}})}{1 + \exp(g(x, t)\theta_{F_{z|x,t,j}})}$
- **Multinomial:** $P(z_j = \ell|x, t; \theta_{F_{z|x,t,j}}) = \frac{\exp(g(x, t)\theta_{F_{z|x,t,j}}^\ell)}{1 + \sum_{i=1}^{L_j-1} \exp(g(x, t)\theta_{F_{z|x,t,j}}^i)}, \ell = 1, \dots, L_j - 1$

A2: Dynamic Likelihood

$$\begin{aligned}
 \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_T} \underbrace{f(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F)}_{\text{Density of } z^c | z^{\neg c}, x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \\
 & \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^N \underbrace{f(z_n^{\neg c} | x_n, t_n; \theta_F)}_{\text{Density of } z^{\neg c} | x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}
 \end{aligned}$$

- Success mostly unobservable for applicants from fewer than six years ago

A2: Heckman (1979)-Corrected Dynamic Likelihood

$$\begin{aligned}
 \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_T} \underbrace{\Phi(x_n, t_n; \theta_H)}_{\text{Selection}} \underbrace{f\left(z_n | x_n, t_n, \frac{\phi(x_n, t_n; \theta_H)}{\Phi(x_n, t_n; \theta_H)}; \theta_F\right)}_{\text{Density of } z|x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A) \mathbb{1}(d_{2,n}=A)}_{\text{Conditional choice probabilities}} \\
 & \underbrace{(1 - P_{d_{2,n}}(\theta_A)) \mathbb{1}(d_{2,n}=R)}_{\text{Conditional choice probability}} \prod_{n=N_T+1}^N \underbrace{(1 - \Phi(x_n, t_n; \theta_H))}_{\text{Selection}} \underbrace{f\left(z_n^{-c} | x_n, t_n, \frac{-\phi(x_n, t_n; \theta_H)}{1 - \Phi(x_n, t_n; \theta_H)}; \theta_F\right)}_{\text{Density of } z^{-c}|x, t} \\
 & \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^N \underbrace{P_{y_n}(\theta_S) \mathbb{1}(y_n=S) (1 - P_{y_n}(\theta_S)) \mathbb{1}(y_n=F)}_{\text{Success probabilities}}
 \end{aligned}$$

- *Committee Score* unobservable for applicants who didn't survive Round 1

A2: Static Model

- **Motivation:** Interpretable cutoff probabilities in both rounds

Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S | t) \mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{P(y = S | x, t, \bar{k}) \mathbb{1}(d_1 = T)}_{\text{If transition, get } P(\text{Success})} + \underbrace{P(y_1^* = S | t) \mathbb{1}(d_1 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

- $P(y = S | x, t, \bar{k})$ estimates inconsistent, but fitted values account for z 's effect on y

A2: Static Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \\ \prod_{n=1}^N \underbrace{[P_{y_n}(\theta_{S_1}) P_{y_n}(\theta_{S_2})]^{\mathbb{1}(y_n=S)} [(1 - P_{y_n}(\theta_{S_1}))(1 - P_{y_n}(\theta_{S_2}))]^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- Success mostly unobservable for applicants from fewer than six years ago

A2: Lagrangian with Matriculation

Portfolio Choice Problem:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N P(y_n = S | w_n) P(m_n = M | w_n) \mathbb{1}(d_n = A),$$

$$\text{such that: } \sum_{n=1}^N P(m_n = M | w_n) \mathbb{1}(d_n = A) \leq N_M$$

$$\mathcal{L}(d_1, \dots, d_N; \lambda) = \sum_{n=1}^N [P(y_n = S | w_n) \mathbb{1}(d_n = A) + \lambda \mathbb{1}(d_n = R)] P(m_n = M | w_n) + \lambda (N_M - \overline{N_M})$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(w_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | w_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S | w_n) = \lambda^* \\ R & \text{if } P(y_n = S | w_n) < \lambda^* \end{cases}$$

A2: Lagrangian with Matriculation (Large N)

Portfolio Choice Problem:

$$\max_{\delta \in \{\delta: \mathbb{R}^K \rightarrow [0,1]\}} \int_{w \in \mathbb{R}^K} P(y = S|w)P(m = M|w)\delta(w)f(w)dw,$$

$$\text{such that: } \int_{w \in \mathbb{R}^K} P(m = M|w)\delta(w)f(w)dw \leq N_M$$

$$\mathcal{L}(\delta; \lambda) = \int_{w \in \mathbb{R}^K} [P(y = S|w)\delta(w) + \lambda(1 - \delta(w))] P(m = M|w)f(w)dw + \lambda(N_M - \overline{N}_M)$$

Cutoff Rule Solution:

$$\forall w \in \mathbb{R}^K, \delta(w|\lambda^*) = \begin{cases} 1 & \text{if } P(y = S|w) > \lambda^* \\ [0, 1] & \text{if } P(y = S|w) = \lambda^* \\ 0 & \text{if } P(y = S|w) < \lambda^* \end{cases}$$

Upshot: Point-identified cutoff probability λ^* is with respect to success probabilities

A2: Lagrangian with Matriculation, Round 2

Portfolio Choice Problem:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S | w_n) P(m_n = M | w_n) \mathbb{1}(d_{2,n} = A),$$

$$\text{such that: } \sum_{n=1}^{N_T} P(m_n = M | w_n) \mathbb{1}(d_{2,n} = A) \leq N_M$$

$$\mathcal{L}(d_{2,1}, \dots, d_{2,N_T}; \lambda_2) = \sum_{n=1}^{N_T} [P(y_n = S | w_n) \mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)] P(m_n = M | w_n) + \lambda_2 (N_M - \overline{N_M})$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N_T\}, d_{2,n}(w_n | \lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S | w_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S | w_n) = \lambda_2^* \\ R & \text{if } P(y_n = S | w_n) < \lambda_2^* \end{cases}$$

A2: Lagrangian with Matriculation, Round 1

Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N EV_{2,n}(x_n; \bar{\lambda}_2^*) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n) = \int_{\tilde{z}_n} \max\{P(y_n = S | x_n, \tilde{z}_n) - \bar{\lambda}_2^*, 0\} P(m_n = M | x_n, \tilde{z}_n) dF_{z|x}(\tilde{z}_n | x_n)$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^* \end{cases}$$

A2: Matriculation-Dynamic Likelihood

Matriculation Probability:

$$P_m = P(m = M | x, z, t; \theta_M) = \frac{\exp(f_M(x, z, t)\theta_M)}{1 + \exp(f_M(x, z, t)\theta_M)}$$

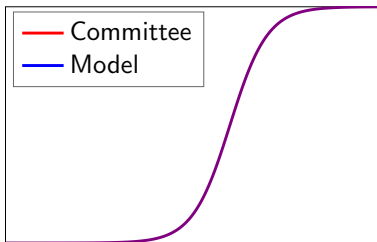
$$\begin{aligned} \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_A} \underbrace{P_{m_n}(\theta_M)^{\mathbb{1}(m_n=M)} (1 - P_{m_n}(\theta_M))^{\mathbb{1}(m_n=D)}}_{\text{Matriculation probabilities}} \\ & \prod_{n=1}^{N_T} \underbrace{f(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F)}_{\text{Density of } z^c | z^{\neg c}, x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \\ & \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^N \underbrace{f(z_n^{\neg c} | x_n, t_n; \theta_F)}_{\text{Density of } z^{\neg c} | x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}} \end{aligned}$$

A2: Empirical Considerations

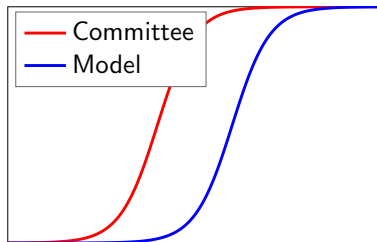
- **Selection Effects:** Track success for all applicants, including non-matriculants
 - In selected samples, success parameters are biased down (e.g. quantitative GRE)
- **Potential Outcomes:** Track which graduate program all applicants attended, and add *Program Rank* as a control variable in success blocks of likelihood
 - In decision blocks containing success, fix *Program Rank* at this university's rank
 - If a variable only predicts success by helping applicant get into elite graduate program, then committee should not use that variable to rank applicants
- **Unobservables:** I may not observe all information that committee observes
 - **Example:** Recommendation letter has nuances that algorithms can't detect
 - Include subjective committee score in z as a proxy for this information

A2: Deviation Gains Procedure

- 1 Write Bellman equations for committee's optimization problem
- 2 Estimate parameters. Rationality assumption distorts estimates
- 3 Re-estimate parameters without rationality, and use to solve model
- 4 Compare selections from model's decision rule to committee's decision rule
- 5 For this comparison, evaluate selections at perturbations of estimates



Optimal Committee



Suboptimal Committee

A2: Deviation Gains Algorithm

Dynamic/Static/Myopic:

- 1 Rank each t 's applicants by $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$ with $\sigma = 0$ for dynamic and myopic, or $P(y = S|x, t, \bar{k})$ for static. Transition top N_T , and reject the rest
- 2 Rank surviving applicants by $P(y = S|x, z, t, \bar{k})$, and accept top N_A survivors.
Compute $\overline{P(y = S|x, z, t, \bar{k})}$ of acceptance set, and compare to committee's set

Matriculation-Dynamic/Static:

- 1 Rank each t 's applicants by $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$ with $\sigma = 0$ for dynamic, or $P(y = S|x, t, \bar{k})$ for static (best analogue). Transition top N_T , and reject the rest
- 2 Rank surviving applicants by $P(y = S|x, z, t, \bar{k})$ and accept them in descending order, each matriculating with probability $P(m = M|x, z, t)$, until N_M matriculate.
Compute $\overline{P(y = S|x, z, t, \bar{k})}$ of matriculation set, and compare to committee's set

A2: Deviation Gains Interpretation

- Under meaningfully-sized deviation gains, committee may have incorrect beliefs about which types of applicants are most likely to succeed
 - Too few observations for a true out-of-sample test, and cannot account for changes in the determinants of success over time, such as writing essays with AI
- Committee may use success-predicting variables that I do not observe
 - Because the success probability measure used to evaluate gains cannot detect such variables, any edge the committee's private information provides goes unmeasured
 - Comprehensive data, especially committee score in z , protects against this limitation

A2: Extensions

- **Extension 1:** Uncover additional variables that predict matriculation but not failure
 - **Example:** Applicants with work experience more likely to succeed and matriculate
 - Can help committee find “people that we can afford” (*Moneyball*)
- **Extension 2:** Relax assumption that committee maximizes $P(\text{Success})$
 - Instead maximize combination of $P(\text{Success})$ and other objectives
 - **Example:** Committee may want diverse cohort (e.g. across fields):
 $P(\text{Success})^\alpha \times \text{Diversity}^{1-\alpha}$, where $\alpha \in [0, 1]$ can be estimated
- **Extension 3:** Waitlist and multiyear models (see [online appendix](#))
 - Three-round models, and two-round models that repeat over infinite horizon
 - States contain info about past rounds/years, and committee values future rounds/years (memory includes matriculation probability and diversity)

A3: Summary Statistics

Objective Variables

	Full Sample		Transition = 1		Accept = 1		Matriculate = 1	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Transition	0.4204	0.4937	1.0000	0.0000	1.0000	0.0000	1.0000	0.0000
Waitlist	0.3324	0.4712	0.7686	0.4220	0.0743	0.2628	0.0896	0.2877
Accept	0.1103	0.3134	0.2625	0.4402	1.0000	0.0000	1.0000	0.0000
Matriculate	0.0339	0.1811	0.0807	0.2725	0.3074	0.4623	1.0000	0.0000
Open House	0.0353	0.1847	0.0817	0.2741	0.3267	0.4702	0.5522	0.5010
Attend Program	0.6024	0.4895	0.6752	0.4685	0.7704	0.4214	1.0000	0.0000
Complete Program	0.5578	0.4968	0.6210	0.4854	0.6654	0.4728	0.6709	0.4729
Top 100 Placement	0.1971	0.3979	0.2472	0.4316	0.3074	0.4623	0.3418	0.4773
Program Rank	88.9133	57.4823	78.5844	57.1361	63.3828	54.1845	37.0000	0.0000
Missing Program Rank	0.4199	0.4937	0.3422	0.4747	0.2490	0.4333	0.0000	0.0000
Placement Rank	127.7505	34.9306	123.0222	37.2570	117.2696	40.5746	117.4430	37.5728
Missing Placement Rank	0.4444	0.4970	0.3830	0.4864	0.3346	0.4728	0.3291	0.4729
2016	0.2052	0.4040	0.1696	0.3754	0.1790	0.3841	0.2025	0.4045
2017	0.2014	0.4011	0.1788	0.3833	0.1829	0.3873	0.1772	0.3843
2018	0.1928	0.3946	0.2686	0.4435	0.2062	0.4054	0.2532	0.4376
2019	0.2027	0.4021	0.2084	0.4064	0.2179	0.4136	0.2152	0.4136
Winsorized Age	25.9910	3.1479	26.2635	3.1823	26.0817	3.0652	26.3291	3.2806
Female	0.3912	0.4881	0.4147	0.4929	0.4397	0.4973	0.4177	0.4963
U.S.	0.2357	0.4245	0.2380	0.4261	0.2996	0.4590	0.3038	0.4628
International Asian	0.4809	0.4997	0.5230	0.4997	0.3424	0.4754	0.4304	0.4983
GRE Verbal Percentile	68.0129	22.5869	76.8421	18.3703	79.5703	18.3047	76.9744	19.4041
GRE Quantitative Percentile	88.4550	11.6330	93.1362	5.4135	93.5391	4.4614	92.3077	5.7554
GRE Analytic Percentile	50.0613	27.5744	56.3653	26.5466	60.3398	26.0880	57.4615	26.6481
Undergraduate GPA	3.6307	0.3026	3.6645	0.3003	3.7227	0.2610	3.7068	0.2880
TOEFL/IELTS	7.4376	0.5587	7.5641	0.5834	7.7011	0.6667	7.4464	0.6137
Elite U.S. University	0.1267	0.3327	0.1736	0.3790	0.2257	0.4188	0.1772	0.3843
Elite U.S. Liberal Arts College	0.0296	0.1696	0.0347	0.1832	0.0428	0.2028	0.0633	0.2450
Elite Foreign University	0.0386	0.1928	0.0490	0.2160	0.0350	0.1842	0.0253	0.1581
Other Foreign University	0.4049	0.4910	0.4188	0.4936	0.3619	0.4815	0.4430	0.4999
Missing Undergraduate Institution	0.2203	0.4145	0.1910	0.3933	0.1984	0.3996	0.1266	0.3346
Graduate Degree	0.5067	0.5001	0.5730	0.4949	0.5136	0.5008	0.5190	0.5028
Work Experience	0.5247	0.4995	0.5894	0.4922	0.6770	0.4685	0.7468	0.4376
#Professor Recommenders	2.6230	0.7371	2.6302	0.7165	2.5992	0.7118	2.5696	0.7104
#Profic Recommenders	1.2602	1.0686	1.3912	1.0840	1.4825	1.0424	1.3038	1.0664
Missing GRE	0.0335	0.1800	0.0102	0.1006	0.0039	0.0624	0.0127	0.1125
Missing Undergraduate GPA	0.7853	0.4107	0.7487	0.4340	0.6770	0.4685	0.6835	0.4681
Missing TOEFL/IELTS	0.6076	0.4884	0.5935	0.4914	0.6615	0.4741	0.6456	0.4814
Missing Winsorized Age	0.0009	0.0293	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Observations	2329		979		257		79	

Subjective Variables

	Full Sample		Transition = 1		Accept = 1		Matriculate = 1	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Committee Score	6.4070	1.8701	6.4070	1.8701	8.5003	0.8151	8.4854	0.7157
#Math Courses	5.7637	1.8154	5.8173	1.7904	5.6992	1.8640	5.6585	1.6219
Advanced Math	0.4860	0.5004	0.4909	0.5007	0.5680	0.4973	0.4651	0.5047
Letters Length/100	8.3474	3.5749	8.5333	3.8263	10.0514	4.2864	9.7088	3.4876
Letters Positive Terms (%)	11.9770	2.2993	11.9976	2.2077	12.2167	1.9255	12.5110	2.0904
Letters Negative Terms (%)	5.7286	1.3427	5.7216	1.3318	5.7446	1.2313	5.8938	1.2871
Letters Standout Terms (%)	0.6241	0.4020	0.6241	0.4029	0.7151	0.4509	0.7077	0.5109
Letters Ability Terms (%)	1.3326	0.6239	1.3457	0.5964	1.3796	0.5620	1.4334	0.6037
Letters Research Terms (%)	3.4837	1.3319	3.5208	1.3423	3.8548	1.1984	3.8395	1.2121
Letters Grindstone Terms (%)	1.8189	0.7734	1.8674	0.7808	1.9975	0.7588	2.1600	0.8093
Letters Teaching Terms (%)	1.7379	0.7710	1.7489	0.7759	1.7275	0.7536	1.8976	0.7577
Letters Communal Terms (%)	0.5946	0.3890	0.6019	0.3954	0.5494	0.3234	0.5630	0.3875
Letters Agentic Terms (%)	0.4768	0.3150	0.4842	0.3229	0.5039	0.3222	0.5597	0.3314
Letters Terms (%) PC1	-0.0579	1.6183	0.0000	1.5768	0.2176	1.3817	0.5697	1.5386
Letters Terms (%) PC2	-0.0271	1.1177	0.0000	1.1446	0.1002	1.1774	0.1699	1.2823
CV Length/100	4.3532	2.4237	4.3982	2.4758	4.9641	2.7175	4.8261	2.3473
CV Coding	2.8321	2.1135	2.7931	2.1000	2.8618	2.1129	2.5122	1.6450
CV Publication/Working Paper	0.2518	0.4346	0.2571	0.4377	0.3659	0.4836	0.3902	0.4939
Essay Length/100	4.5099	1.5741	4.5173	1.5805	4.7488	1.6373	4.9816	1.7932
Essay Specific Professor	0.4822	0.5003	0.5219	0.5003	0.5691	0.4972	0.6512	0.4822
Essay Economics Terms (%)	9.7284	2.6219	9.7216	2.6635	9.5303	2.7324	9.8941	3.2608
Essay Research Terms (%)	5.7975	2.0501	5.7402	2.0168	5.7313	1.8797	5.5231	1.8432
Essay Positive Terms (%)	11.8293	2.6518	11.7324	2.6650	11.4734	2.4592	11.3169	2.4455
Essay Negative Terms (%)	6.2844	1.9056	6.3233	1.9712	6.2889	2.0860	6.2651	1.9436
Missing Committee Score	0.5844	0.4929	0.0112	0.1055	0.0195	0.1384	0.0633	0.2450
Missing Application Pdf	0.8154	0.3881	0.6650	0.4722	0.5136	0.5008	0.4557	0.5012
Missing #Math Courses	0.8201	0.3842	0.6701	0.4704	0.5214	0.5005	0.4810	0.5028
Missing Letters	0.8175	0.3863	0.6691	0.4708	0.5136	0.5008	0.4557	0.5012
Missing CV	0.8210	0.3835	0.6742	0.4689	0.5214	0.5005	0.4810	0.5028
Missing Essay	0.8192	0.3849	0.6731	0.4693	0.5214	0.5005	0.4557	0.5012
Observations	2329		979		257		79	

A3: Recommendation Letters Principal Components

	PC1	PC2
Positive	0.5044	-0.1769
Negative	0.1479	0.3759
Standout	0.2115	-0.5082
Ability	0.4667	-0.0177
Research	0.1214	0.6418
Grindstone	0.3879	0.3075
Teaching	0.3708	0.0743
Communal	0.1284	-0.1983
Agentic	0.3768	-0.1324
Proportion of Variance	0.2763	0.1456

A3: Logistic Regression Results

Year	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042	2043	2044	2045	2046	2047	2048	2049	2050	2051	2052	2053	2054	2055	2056	2057	2058	2059	2060	2061	2062	2063	2064	2065	2066	2067	2068	2069	2070	2071	2072	2073	2074	2075	2076	2077	2078	2079	2080	2081	2082	2083	2084	2085	2086	2087	2088	2089	2090	2091	2092	2093	2094	2095	2096	2097	2098	2099	2100	2101	2102	2103	2104	2105	2106	2107	2108	2109	2110	2111	2112	2113	2114	2115	2116	2117	2118	2119	2120	2121	2122	2123	2124	2125	2126	2127	2128	2129	2130	2131	2132	2133	2134	2135	2136	2137	2138	2139	2140	2141	2142	2143	2144	2145	2146	2147	2148	2149	2150	2151	2152	2153	2154	2155	2156	2157	2158	2159	2160	2161	2162	2163	2164	2165	2166	2167	2168	2169	2170	2171	2172	2173	2174	2175	2176	2177	2178	2179	2180	2181	2182	2183	2184	2185	2186	2187	2188	2189	2190	2191	2192	2193	2194	2195	2196	2197	2198	2199	2200	2201	2202	2203	2204	2205	2206	2207	2208	2209	2210	2211	2212	2213	2214	2215	2216	2217	2218	2219	2220	2221	2222	2223	2224	2225	2226	2227	2228	2229	2230	2231	2232	2233	2234	2235	2236	2237	2238	2239	2240	2241	2242	2243	2244	2245	2246	2247	2248	2249	2250	2251	2252	2253	2254	2255	2256	2257	2258	2259	2260	2261	2262	2263	2264	2265	2266	2267	2268	2269	2270	2271	2272	2273	2274	2275	2276	2277	2278	2279	2280	2281	2282	2283	2284	2285	2286	2287	2288	2289	2290	2291	2292	2293	2294	2295	2296	2297	2298	2299	2300	2301	2302	2303	2304	2305	2306	2307	2308	2309	2310	2311	2312	2313	2314	2315	2316	2317	2318	2319	2320	2321	2322	2323	2324	2325	2326	2327	2328	2329	2330	2331	2332	2333	2334	2335	2336	2337	2338	2339	2340	2341	2342	2343	2344	2345	2346	2347	2348	2349	2350	2351	2352	2353	2354	2355	2356	2357	2358	2359	2360	2361	2362	2363	2364	2365	2366	2367	2368	2369	2370	2371	2372	2373	2374	2375	2376	2377	2378	2379	2380	2381	2382	2383	2384	2385	2386	2387	2388	2389	2390	2391	2392	2393	2394	2395	2396	2397	2398	2399	2400	2401	2402	2403	2404	2405	2406	2407	2408	2409	2410	2411	2412	2413	2414	2415	2416	2417	2418	2419	2420	2421	2422	2423	2424	2425	2426	2427	2428	2429	2430	2431	2432	2433	2434	2435	2436	2437	2438	2439	2440	2441	2442	2443	2444	2445	2446	2447	2448	2449	2450	2451	2452	2453	2454	2455	2456	2457	2458	2459	2460	2461	2462	2463	2464	2465	2466	2467	2468	2469	2470	2471	2472	2473	2474	2475	2476	2477	2478	2479	2480	2481	2482	2483	2484	2485	2486	2487	2488	2489	2490	2491	2492	2493	2494	2495	2496	2497	2498	2499	2500
2019	0.3628	0.3628	0.2671	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0.0220	0																																																																																		

- **Accept ($p < .1$):** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Quantitative Percentile (+), Work Experience (+), Committee Score (+), CV Coding (-)
- **Success ($p < .1$):** Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Essay Positive Terms (+), Program Rank (-), Missing Program Rank (-)

A3: Logistic Regression AMEs

Dependent Variable	Age	Age ²	Married	12th Percentile	12th Percentile	12th Percentile	Program Rank	Program Rank	Program Rank
2004	-0.0211	-0.0014	-0.0307	0.0020	0.0040	0.0122	0.0001	0.0001	-0.0018
2007	-0.0111	-0.0044	-0.0070	-0.0021	-0.0021	-0.0040	-0.0001	-0.0001	-0.0001
2010	-0.0040	-0.0044	-0.0011	0.0011	0.0011	0.0011	0.0011	0.0011	0.0011
2014	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011
2018	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011
Winsorized Age	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011
Female	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011
U.S.	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011
International Asian	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011
GRE Quantitative Percentile	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011
Work Experience	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011
Committee Score	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011
CV Coding	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011
Undergraduate GPA	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011
TOEFL/IELTS	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011
Essay Positive	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011
Program Rank	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011
Missing Program Rank	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011	-0.0011
Sample	50	50	Average = 1	50	50	50	50	50	50
Observations	2004	2007	2010	2014	2018	2019	2020	2021	2022

- Accept ($p < .1$):** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Quantitative Percentile (+), Work Experience (+), Committee Score (+), CV Coding (-)
- Success ($p < .1$):** Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Essay Positive Terms (+), Program Rank (-), Missing Program Rank (-)

A3: Logistic Regression Results (Synthetic)

Student Variable	Accept	Reject	Missing	TOEFL PctRank	TOEFL PctRank	TOEFL PctRank	Program Rank	Program Rank	Program Rank
2016	0.2081	0.1213	0.1276	0.0887	0.0822	0.0220	0.0822	0.0822	1.1027
	(0.2081)	(0.1213)	(0.1276)	(0.0887)	(0.0822)	(0.0220)	(0.0822)	(0.0822)	(1.1027)
2017	0.1398	0.1807	0.0910	0.0207	0.2061	0.0008	0.1808**	0.2020***	0.3019
	(0.1398)	(0.1807)	(0.0910)	(0.0207)	(0.2061)	(0.0008)	(0.1808)	(0.2020)	(0.3019)
2018	0.0272	0.0292	0.0447	0.0241	0.1762	0.1070	0.0272	0.0272	0.0292
	(0.0272)	(0.0292)	(0.0447)	(0.0241)	(0.1762)	(0.1070)	(0.0272)	(0.0272)	(0.0292)
2019	0.1284	0.1232	0.0107	0.0000	0.1072	0.1072	0.1284	0.1284	0.1284
	(0.1284)	(0.1232)	(0.0107)	(0.0000)	(0.1072)	(0.1072)	(0.1284)	(0.1284)	(0.1284)
2020	0.1141	0.0871	0.0176	0.0000	0.1176	0.0000	0.1141	0.1141	0.1141
	(0.1141)	(0.0871)	(0.0176)	(0.0000)	(0.1176)	(0.0000)	(0.1141)	(0.1141)	(0.1141)
Missing Age	0.2021	0.2490	0.0498	0.0107	0.0103	0.2062	0.0107	0.0107	0.0107
	(0.2021)	(0.2490)	(0.0498)	(0.0107)	(0.0103)	(0.2062)	(0.0107)	(0.0107)	(0.0107)
Gender	0.0220	0.0242	0.0402	0.0170	0.0170	0.0100	0.0220	0.0220	0.0220
	(0.0220)	(0.0242)	(0.0402)	(0.0170)	(0.0170)	(0.0100)	(0.0220)	(0.0220)	(0.0220)
Books	0.1020	0.1020	0.0000	0.0000	0.1020	0.1020	0.1020	0.1020	0.1020
	(0.1020)	(0.1020)	(0.0000)	(0.0000)	(0.1020)	(0.1020)	(0.1020)	(0.1020)	(0.1020)
US	0.1000	0.1000	0.0000	0.1000	0.1000	0.1000	0.1000	0.1000	0.1000
	(0.1000)	(0.1000)	(0.0000)	(0.1000)	(0.1000)	(0.1000)	(0.1000)	(0.1000)	(0.1000)
International Rank	0.1282	0.1282	0.0000	0.1282	0.1282	0.1282	0.1282	0.1282	0.1282
	(0.1282)	(0.1282)	(0.0000)	(0.1282)	(0.1282)	(0.1282)	(0.1282)	(0.1282)	(0.1282)
GRE Verbal Percentile	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
GRE Quantitative Percentile	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Undergraduate GPA	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
TOEFL iBT/ETS	0.1209	0.1276	0.0075	0.0000	0.0000	0.0000	0.1209	0.1209	0.1209
	(0.1209)	(0.1276)	(0.0075)	(0.0000)	(0.0000)	(0.0000)	(0.1209)	(0.1209)	(0.1209)
Elite U.S. University	0.0471	0.1007	0.2066	0.1100	0.1100	0.1100	0.0471	0.0471	0.0471
	(0.0471)	(0.1007)	(0.2066)	(0.1100)	(0.1100)	(0.1100)	(0.0471)	(0.0471)	(0.0471)
Elite U.S. Liberal Arts College	0.0462	0.0462	0.0000	0.0000	0.0462	0.0462	0.0462	0.0462	0.0462
	(0.0462)	(0.0462)	(0.0000)	(0.0000)	(0.0462)	(0.0462)	(0.0462)	(0.0462)	(0.0462)
Other Foreign University	0.1797	0.1797	0.0000	0.1797	0.1797	0.1797	0.1797	0.1797	0.1797
	(0.1797)	(0.1797)	(0.0000)	(0.1797)	(0.1797)	(0.1797)	(0.1797)	(0.1797)	(0.1797)
Missing Undergraduate Institution	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Graduate Degree	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Work Experience	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
gPA/Undergraduate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
gPA/Graduate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Missing GRE	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Missing TOEFL iBT/ETS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Commitment Score	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
gMath Courses	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Advanced Math	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Letters Length/100	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Letters Terms (%) PC1	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Letters Terms (%) PC2	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
CV Length/100	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
CV Coding	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
CV Publications/Working Paper	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Essay Length/100	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Essay Specific/100	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Essay Economics Terms (%)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Essay Research Terms (%)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Essay Politics Terms (%)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Essay Negative Terms (%)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Missing Undergraduate PAF	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Open House	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Program Rank	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Missing Program Rank	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)

- **Accept ($p < .1$):** 2017 (-), Winsorized Age (-), Letters Terms (%) PC2 (-), Missing Committee Score (-)
- **Success ($p < .1$):** Female (+), Elite U.S. University (+), Missing TOEFL/IELTS (+), #Math Courses (-), Letters Length/100 (-), CV Coding (+), Missing Committee Score (+), Program Rank (-), Missing Program Rank (-)

A3: Logistic Regression AMEs (Synthetic)

Dependent Variable	Accept	Success	Math	TOEFL	IELTS	Program Rank	Program Rank	Program Rank
2014	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
2017	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
2018	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
2019	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Western Age	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Female	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
U.S.	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
International Age	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
USC Virtual Presence	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
USC Quantitative Presence	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
USC Analytic Presence	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Undergraduate GPA	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
TOEFL/IELTS	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Elite U.S. University	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Elite U.S. Liberal Arts College	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Elite Foreign University	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Elite Foreign University	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Missing Undergraduate Institution	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Graduate Degree	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Work Experience	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
JPProfile Recommenders	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
JPProfile Recommenders	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Missing GRE	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Missing Undergraduate GPA	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Missing TOEFL/IELTS	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Quantitative Score	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Math Courses	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Advanced Math	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Letter Length/100	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Letter Score [%] PC2	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Letter Score [%] PC2	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
CV Length/100	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
CV Coding	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
CV Publications/Working Paper	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Essay Length/100	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Essay Specific Profound	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Essay Economic Terms [%]	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Essay Positive Terms [%]	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Essay Negative Terms [%]	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Missing Covariate Score	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Missing Application PM	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Open Hours	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Program Rank	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Missing Program Rank	-0.0001	-0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001

- **Accept** ($p < .1$): 2017 (-), Winsorized Age (-), Letters Terms (%) PC2 (-), Missing Committee Score (-)
- **Success** ($p < .1$): Female (+), Elite U.S. University (+), Missing TOEFL/IELTS (+), #Math Courses (-), Letters Length/100 (-), CV Coding (+), Missing Committee Score (+), Program Rank (-), Missing Program Rank (-)

A3: Lasso Logistic Regression Results

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.1107								
2017	-0.4928								
2018	-2.4456	0.0186							
2019	-1.7752								
Winsorized Age				0.0084	0.0047	0.0184			
Female	0.2674	0.5589							
U.S.	-0.1959						0.0477		
International Asian	-0.8686	-0.8651	0.1271			-0.5295	-0.7927		
GRE Verbal Percentile	0.0227	0.0047	-0.0015	0.0015		-0.0085			
GRE Quantitative Percentile	0.1027	0.0174	-0.0273	0.0179	0.0125	-0.2706	-0.2243	-0.0070	
GRE Analytic Percentile	0.0056	0.0035							
Undergraduate GPA	0.9812	0.7135				-0.1316		0.2561	
TOEFL/IELTS	0.3055		-0.2943	0.0300		0.0934	-0.6903	-0.5850	-2.2692
Elite U.S. Liberal Arts College	0.4024	0.1112							
Elite Foreign University	-0.4025								
Other Foreign University	0.0790	0.0940	0.1998					-0.2265	
Missing Undergraduate Institution		-0.0036							
Graduate Degree	0.1178								
Work Experience	0.6006	0.4068	0.0404	0.0961	0.0470	0.1976		-2.0809	0.0922
#Prolific Recommenders	0.1534	0.0301	-0.0870	0.0526	0.0363		-0.0789	-0.0358	
Missing GRE	-0.6534			-0.0697					
Missing Undergraduate GPA	-0.2348	-0.1148							
Missing TOEFL/IELTS	0.1447			-0.1243	-0.1295	1.8402	2.2548	-0.4899	
Committee Score		3.4243			0.0353				
#Math Courses	-0.0553	0.0354							
Advanced Math	0.4302	0.7352							
Letters Length/100	0.1196	0.0587							
Letters Terms (%) PC1	0.0746	0.1485	0.1258			0.0102			
Letters Terms (%) PC2	-0.0162	-0.2901							
CV Length/100	0.0219	0.0298		0.0014		-0.0286	-0.1179		
CV Coding	-0.0038	-0.1754		0.0165	0.0076	-0.2540	-0.4154	-0.0998	
CV Publication/Working Paper	0.5563	0.6436		0.0892		-0.0687			
Essay Length/100	0.1202	0.0295							
Essay Specific Professor	0.3403	0.3958	0.0674						
Essay Economics Terms (%)	-0.0016			-0.0405	-0.0325	0.4473	0.6570	0.2117	
Essay Research Terms (%)		-0.0278					-0.1506		
Essay Positive Terms (%)	-0.0049	0.0382		0.0192	0.0140	0.0267			
Essay Negative Terms (%)		-0.0417							
Missing Committee Score		-0.8416			-0.2615	-0.0714	4.0146	0.7071	
Missing Application Pdf	-1.0365								
Open House			1.0982						
Program Rank						-0.0082		0.1385	
Missing Program Rank						-2.2557		19.4273	
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	257	2329	2329	2329	2329	2329	2329
Pseudo-R ²	0.2841	0.8007	0.1281	0.0196	0.0213	0.2265	0.0206	0.0278	0.2693
CV Pseudo-R ²	0.2441	0.7605	0.0596	0.0062	0.0129	0.2190	0.0121	0.0183	0.2638

- Accept:** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Verbal Percentile (+), GRE Quantitative Percentile (+), GRE Analytic Percentile (+), Undergraduate GPA (+), Elite U.S. Liberal Arts College (+), Elite Foreign University (-), Other Foreign University (+), Work Experience (+), # Prolific Recommenders (+), Missing Undergraduate GPA (-), Committee Score (+), #Math Courses (-), Advanced Math (+), Letters Length/100 (+), Letters Terms (%) PC1 (+), Letters Terms (%) PC2 (-), CV Length/100 (+), CV Coding (-), CV Publication/Working Paper (+), Essay Length/100 (+), Essay Specific Professor (+), Economics Research Terms (%) (-), Essay Positive Terms (%) (-), Essay Negative Terms (%) (-), Missing Committee Score (-)
- Matriculate:** 2018 (+), International Asian (+), GRE Verbal Percentile (-), GRE Quantitative Percentile (-), TOEFL/IELTS (-), Other Foreign University (+), Missing Undergraduate Institution (-), Work Experience (+), #Prolific Recommenders (-), Letters Terms (%) PC1 (+), Essay Specific Professor (+), Open House (+)
- T100 Placement:** Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Letters Terms (%) PC1 (+), Essay Economics Terms (%) (-), Essay Positive Terms (%) (+), Missing Committee Score (-), Program Rank (-), Missing Program Rank (-)

Restricted

[illegible]

A3: Restricted Static Estimates

	Round 1 Cutoff	Round 2 Cutoff	Sigma
2015	29.36% (23.20%, 36.38%)	39.20% (32.75%, 46.05%)	
⋮	⋮	⋮	
2019	34.04% (28.67%, 39.85%)	39.62% (32.54%, 47.16%)	
Sigma			0.0064 (0.0035)

- Static model has interpretable Round 1 cutoff probabilities
- Round 2 cutoff probabilities always greater than those in Round 1

A3: Myopic Estimates

Unrestricted

[illegible]

	(S. 0001)			
Obs./Param./LL/AC	2120	297	16470.7	29136.6

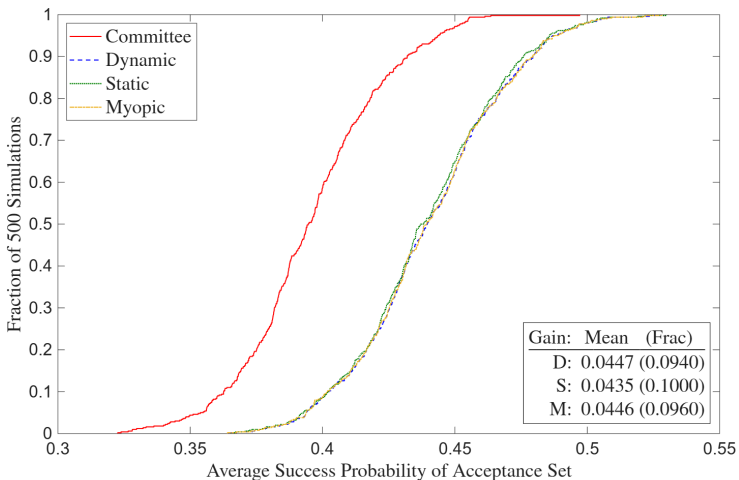
Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Restricted

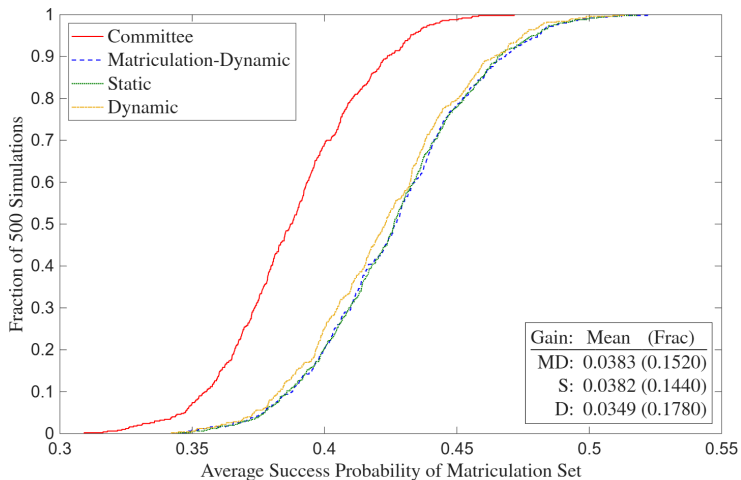
	Consistent Score	Case1/1	Case2/2	Case3/3	Source
Age	1.077*** (0.000)	0.677*** (0.000)	0.576 (0.018, 0.948)	0.822*** (0.000)	0.269** (0.007)
2018	0.979 (0.000)	0.609 (0.000)	0.792 (0.000)	0.808 (0.000)	0.834 (0.000)
2019	0.986 (0.000)	0.710 (0.000)	0.871 (0.000)	0.810 (0.000)	0.875 (0.000)
2020	0.719*** (0.000)	0.507 (0.000)	0.495 (0.000)	0.814 (0.000)	0.588 (0.000)
2021	0.719*** (0.000)	0.498 (0.000)	0.495 (0.000)	0.810 (0.000)	0.588 (0.000)
2022	1.077*** (0.000)	0.609 (0.000)	0.792 (0.000)	0.808 (0.000)	0.834 (0.000)
2023	0.889*** (0.000)	0.609 (0.000)	0.792 (0.000)	0.808 (0.000)	0.834 (0.000)
Missing Age					
				(0.000)	0.834 (0.000)
Female				(0.000)	0.834 (0.000)
U.S.				(0.000)	0.834 (0.000)
International				(0.000)	0.834 (0.000)
GRE Verbal Score				(0.000)	0.834 (0.000)
GRE Quantitative Score				(0.000)	0.834 (0.000)
GRE Analytical Score				(0.000)	0.834 (0.000)
TOEFL iBT				(0.000)	0.834 (0.000)
Non U.S. Liberal Arts College				(0.000)	0.834 (0.000)
Non Foreign University				(0.000)	0.834 (0.000)
Other Foreign University				(0.000)	0.834 (0.000)
Missing Undergraduate Institution				(0.000)	0.834 (0.000)
Undergraduate Degree				(0.000)	0.834 (0.000)
Major				(0.000)	0.834 (0.000)
Psychology Recommendations				(0.000)	0.834 (0.000)
Physics Recommendations				(0.000)	0.834 (0.000)
Missing GRE				(0.000)	0.834 (0.000)
Missing TOEFL iBT				(0.000)	0.834 (0.000)
Black/Asian	0.007 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Adjusted Math	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-200	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-100	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-50	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-25	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-10	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-5	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-2	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-1	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.5	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.25	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.1	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.05	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.01	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000000000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000000000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000000000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000000000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000000000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000000000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000000000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000000000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000000000000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000000000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000000000000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000000000000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000000000000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000000000000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000000000000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000000000000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000000000000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000000000000000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000000000000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000000000000000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000000000000000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000000000000000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000000000000000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000000000000000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000000000000000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000000000000000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000000000000000000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000000000000000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000000000000000000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000000000000000000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000000000000000000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000000000000000000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000000000000000000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000000000000000000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000000000000000000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000000000000000000000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000000000000000000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000000000000000000000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000000000000000000000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000000000000000000000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000000000000000000000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000000000000000000000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.00000000000000000000000001	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000000000000000000000005	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.0000000000000000000000000025	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)	0.001 (0.000)
Latent Length-0.000000000000000000000000001	0.001				

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Confidence intervals are 95%.

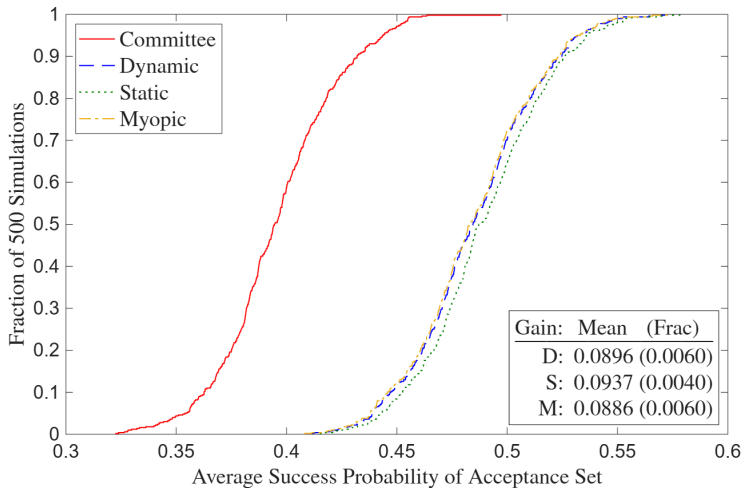
A3: Full Deviation Gains (Acceptance)



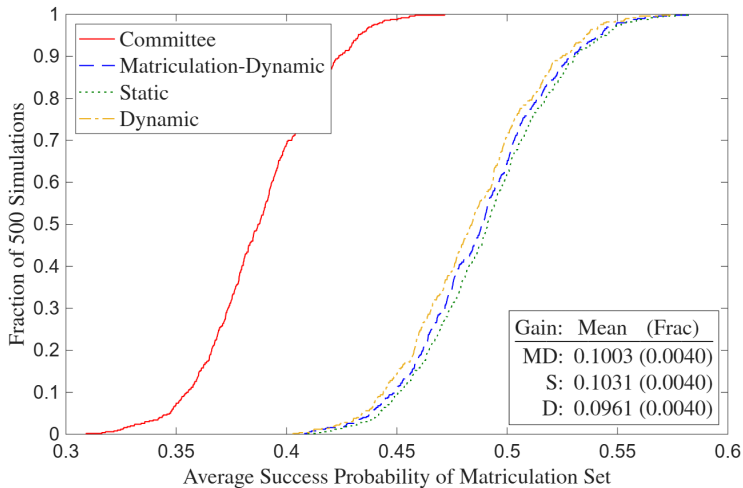
A3: Full Deviation Gains (Matriculation)



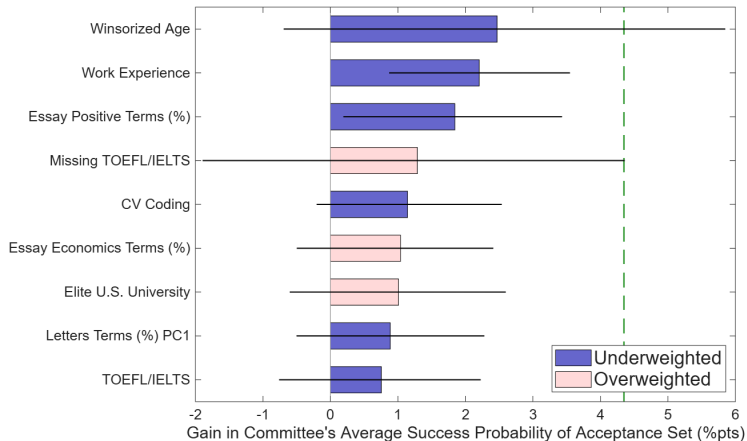
A3: Uncorrected Deviation Gains (Acceptance)



A3: Uncorrected Deviation Gains (Matriculation)

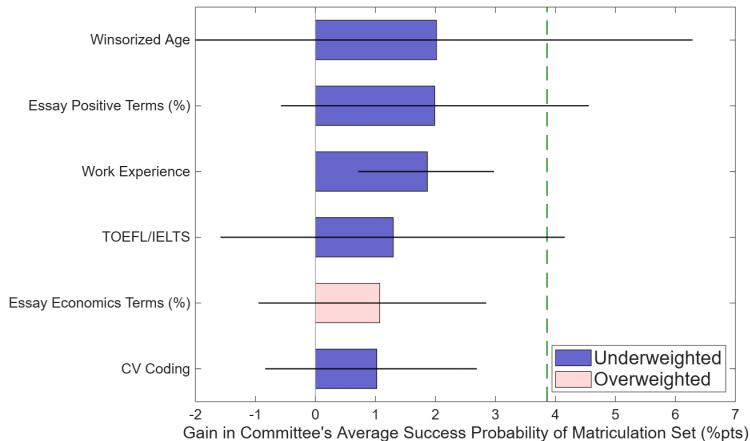


A3: Full Deviation Gains Attribution (Acceptance)



- 1 Experience
- 2 English Proficiency
- 3 Essay/Letters Subtext
- 4 Industry Background
- 5 Undergraduate

A3: Full Deviation Gains Attribution (Matriculation)



- 1 Experience
- 2 English Proficiency
- 3 Essay Subtext
- 4 Industry Background

A3: Static Attribution

	Gain (%pt)	Sample Mean	Committee Mean	1-Variable Optimal Mean	Joint Optimal Mean	1-Variable Difference (SD)	Joint Difference (SD)
Winsorized Age	2.4692	25.9910	26.2840	29.9572	29.5331	1.1674	1.0326
Work Experience	2.2037	0.5247	0.7276	0.9844	0.9650	0.5141	0.4752
Essay Positive Terms (%)	1.8431	11.7499	11.6521	12.5376	12.1183	0.7858	0.4137
Missing TOEFL/IELTS	1.2872	0.6076	0.6654	0.0739	0.4086	-1.2110	-0.5258
CV Coding	1.1426	2.8001	2.7820	3.3399	2.9415	0.6244	0.1785
Essay Economics Terms (%)	1.0382	9.7228	9.7423	9.0284	9.5388	-0.6411	-0.1828
Elite U.S. University	1.0086	0.1267	0.2335	0.0078	0.0389	-0.6784	-0.5848
Letters Terms (%) PC1	0.8868	-0.0106	0.1374	0.4969	0.1387	0.5202	0.0018
TOEFL/IELTS	0.7541	7.4376	7.5675	7.7759	7.6827	0.5957	0.3294

» Deviation Gains Attribution

A3: Matriculation-Static Attribution

	Gain (%pt)	Sample Mean	Committee Mean	1-Variable Optimal Mean	Joint Optimal Mean	1-Variable Difference (SD)	Joint Difference (SD)
Winsorized Age	2.0176	25.9910	26.3284	30.9159	29.7737	1.4580	1.0950
Essay Positive Terms (%)	1.9874	11.7499	11.6107	12.9136	12.2763	1.1534	0.5905
Work Experience	1.8659	0.5247	0.7938	0.9949	0.9799	0.4077	0.3725
TOEFL/IELTS	1.2973	7.4376	7.4939	7.8501	7.6128	1.0238	0.3397
Essay Economics Terms (%)	1.0726	9.7228	9.9994	9.0856	9.5738	-0.8271	-0.3821
CV Coding	1.0219	2.8001	2.6661	3.3395	2.8782	0.7618	0.2374

» Deviation Gains Attribution (Matriculation)

A3: Deviation Gains Robustness

	Acceptance Set	Matriculation Set
Observed Gain, Uncorrected	0.1015	0.1124
Observed Gain, Corrected	0.0514	0.0475
Perturbed Gain, Uncorrected	0.0937	0.1031
Perturbed Gain, Corrected	0.0435	0.0382
Likelihood-Ratio P-Value	0.1166	0.1166
Calibrated Signal Scale	0.4197	0.4197
Calibrated In-Sample Gain	0.1242	0.1481
Calibrated True Gain	0.0491	0.0560

» Deviation Gains (Acceptance)

» Deviation Gains (Matriculation)

- 1 Scale down (x, z) parameters with values of scaling factor τ from 0 to 1
- 2 Simulate *Success*, and find specific τ yielding observed joint LR test statistic
- 3 Calculate true deviation gain for that τ

A4: Graduate School Means by Program

	Economics	Government	History	Linguistics	Psychology
Accept	0.1121	0.0430	0.0986	0.0678	0.0535
Covid Year	0.2478	0.2792	0.3029	0.3460	0.3004
Winsorized Age	26.2782	27.9633	26.8478	28.0692	25.8230
Female	0.4201	0.4182	0.4232	0.6133	0.7490
U.S. African/Hispanic/Native	0.0289	0.0766	0.1159	0.0665	0.1728
U.S. Asian	0.0313	0.0421	0.0304	0.0353	0.0576
International Asian	0.5322	0.2510	0.1058	0.2714	0.1523
International Other	0.2517	0.2967	0.2203	0.3256	0.1584
GRE Verbal Percentile	71.8677	81.3486	82.6054	73.5451	75.6192
GRE Quantitative Percentile	86.2163	63.3428	49.3295	57.8455	58.0000
GRE Analytic Percentile	52.6351	72.1905	75.0460	63.6824	69.2077
Undergraduate GPA	3.6534	3.5675	3.6531	3.6567	3.5695
TOEFL/IELTS	7.5232	7.5422	7.4252	7.5444	7.3158
Elite U.S. University	0.1516	0.1847	0.2565	0.1954	0.2840
Elite U.S. Liberal Arts College	0.0423	0.0672	0.0696	0.0231	0.0535
Elite Foreign University	0.0717	0.0578	0.0478	0.0488	0.0206
Other Foreign University	0.5736	0.3953	0.2333	0.5102	0.1852
Graduate Degree	0.7478	0.7553	0.6652	0.7544	0.4259
Work Experience	0.5938	0.7463	0.6638	0.7544	0.7593
#Professor Recommenders	2.5173	2.3469	2.6261	2.4355	2.1831
#Prolific Recommenders	1.0760	0.6912	0.5043	0.7069	0.3128
#Math Courses	6.6396				
Advanced Math	0.4827				
Missing GRE	0.0255	0.4612	0.6217	0.6839	0.4650
Missing Undergraduate GPA	0.6607	0.4827	0.3072	0.5726	0.2490
Missing TOEFL/IELTS	0.5958	0.7714	0.8449	0.7707	0.8827
Observations	2078	2231	690	737	486

- **Economics Highest:** Accept, International Asian, GRE Quantitative Percentile, Elite Foreign University, Other Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, U.S. African/Hispanic/Native, GRE Verbal Percentile, GRE Analytic Percentile, Elite U.S. University, Work Experience, Missing GRE, Missing TOEFL/IELTS

A4: Graduate School Means by Program, Accept = 1

	Economics	Government	History	Linguistics	Psychology
Covid Year	0.1888	0.2188	0.2059	0.2200	0.2308
Winsorized Age	25.0815	27.7292	27.2794	27.1000	25.7692
Female	0.4850	0.5000	0.5588	0.5800	0.6154
U.S. African/Hispanic/Native	0.0386	0.1458	0.1765	0.0400	0.0769
U.S. Asian	0.0429	0.0312	0.0147	0.0600	0.1154
International Asian	0.4163	0.1667	0.0882	0.3600	0.0769
International Other	0.2575	0.2292	0.3235	0.2000	0.1538
GRE Verbal Percentile	84.3980	88.8725	86.6328	77.9161	85.7729
GRE Quantitative Percentile	91.9289	68.3708	54.8113	64.5504	62.3462
GRE Analytic Percentile	69.0355	82.7222	78.0879	68.8958	74.6169
Undergraduate GPA	3.7336	3.6404	3.7204	3.7004	3.6341
TOEFL/IELTS	7.6419	7.6050	7.4730	7.6237	7.3178
Elite U.S. University	0.2833	0.3125	0.3382	0.2200	0.5385
Elite U.S. Liberal Arts College	0.1030	0.1458	0.1176	0.0200	0.1154
Elite Foreign University	0.1030	0.0521	0.0588	0.1000	0.0385
Other Foreign University	0.3863	0.2396	0.2647	0.4000	0.1154
Graduate Degree	0.5837	0.6771	0.7206	0.7400	0.4615
Work Experience	0.6309	0.8125	0.7647	0.7000	0.9231
#Professor Recommenders	2.5494	2.3750	2.8235	2.7200	2.2308
#Prolific Recommenders	1.2833	0.9167	0.6765	1.0800	0.4615
#Math Courses	7.2575				
Advanced Math	0.6781				
Missing GRE	0.0086	0.2917	0.6324	0.5800	0.1923
Missing Undergraduate GPA	0.4893	0.3125	0.3088	0.4800	0.1923
Missing TOEFL/IELTS	0.6567	0.8854	0.8529	0.7600	0.8846
Observations	233	96	68	50	26

- **Economics Highest:** International Asian, GRE Quantitative Percentile, Undergraduate GPA, TOEFL/IELTS, Elite Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, Winsorized Age, Female, U.S. African/Hispanic/Native, Work Experience, Missing GRE, Missing TOEFL/IELTS

A4: Graduate School Correlation Matrix

	Covid	Age	Fem	USAHN	USAsn	IntAsn	IntOth	GREV	GREQ	GREa	GPA	T/I	EUSUni	EUSLA	EFor	OthFor	Grad	Work	Prof	Pro	MGRE	MGPA	MT/I
Covid Year	1.00																						
Winsorized Age	0.01	1.00																					
Female	0.02	-0.13	1.00																				
U.S. African/Hispanic/Native	0.00	-0.04	0.04	1.00																			
U.S. Asian	-0.00	-0.06	0.05	-0.05	1.00																		
International Asian	-0.05	-0.07	0.02	-0.19	-0.14	1.00																	
International Other	0.01	0.18	-0.06	-0.17	-0.12	-0.42	1.00																
GRE Verbal Percentile	0.03	-0.06	-0.04	0.00	0.04	-0.12	-0.15	1.00															
GRE Quantitative Percentile	-0.02	-0.16	-0.10	-0.18	0.00	0.44	-0.10	0.16	1.00														
GRE Analytic Percentile	0.05	-0.06	0.03	0.08	0.08	-0.37	-0.08	0.60	-0.20	1.00													
Undergraduate GPA	-0.02	-0.29	0.07	-0.13	-0.02	0.02	0.00	0.06	0.12	0.05	1.00												
TOEFL/IELTS	-0.01	-0.07	0.01	-0.02	0.01	0.04	0.00	0.24	0.12	0.19	0.00	1.00											
Elite U.S. University	0.01	-0.12	0.02	0.12	0.14	-0.12	-0.22	0.16	-0.00	0.19	-0.01	-0.02	1.00										
Elite U.S. Liberal Arts College	-0.02	-0.02	0.04	0.04	0.04	-0.08	-0.09	0.12	-0.01	0.13	-0.02	-0.00	-0.11	1.00									
Elite Foreign University	-0.00	-0.04	-0.00	-0.06	-0.04	0.09	0.09	0.02	0.08	-0.02	0.01	0.04	-0.12	-0.06	1.00								
Other Foreign University	-0.02	0.20	-0.04	-0.23	-0.14	0.34	0.39	-0.26	0.20	-0.39	0.02	0.02	-0.43	-0.21	-0.22	1.00							
Graduate Degree	0.00	0.41	-0.08	-0.13	-0.09	0.21	0.15	-0.14	0.05	-0.20	-0.22	0.03	-0.21	-0.10	0.05	0.37	1.00						
Work Experience	0.01	0.33	0.05	0.05	0.01	-0.18	0.11	0.01	-0.15	0.11	-0.08	-0.03	-0.02	0.03	-0.03	-0.01	0.10	1.00					
#Professor Recommenders	-0.02	-0.19	-0.04	-0.04	0.01	0.11	-0.09	0.01	0.09	-0.04	0.14	0.03	-0.02	0.02	-0.00	-0.02	-0.03	-0.18	1.00				
#Prolific Recommenders	0.01	-0.06	-0.02	-0.07	-0.01	0.24	-0.10	-0.01	0.25	-0.11	0.04	0.10	-0.01	-0.01	0.06	0.08	0.17	-0.09	0.19	1.00			
Missing GRE	0.09	0.15	0.05	0.11	-0.01	-0.22	0.17	0.09	-0.33	0.17	-0.09	-0.10	-0.03	-0.04	-0.03	-0.01	0.02	0.10	-0.08	-0.20	1.00		
Missing Undergraduate GPA	-0.01	0.18	-0.04	-0.24	-0.15	0.36	0.42	-0.24	0.21	-0.38	0.02	0.04	-0.46	-0.23	0.23	0.83	0.37	-0.02	-0.03	0.10	-0.01	1.00	
Missing TOEFL/IELTS	0.05	-0.07	0.03	0.16	0.11	-0.32	-0.22	0.16	-0.24	0.32	-0.02	-0.02	0.29	0.14	-0.04	-0.58	-0.20	0.05	-0.07	-0.05	0.12	-0.58	1.00
Observations	6222																						

► Insufficiency of Comparing AMEs

A4: Graduate School Logistic Regression Results

Dependent Variable ← Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3966** (0.2010)	-0.4332* (0.2612)	-0.5752* (0.3479)	-0.5501 (0.4129)	-0.1766 (0.5545)
Winsorized Age	-0.0095*** (0.0358)	0.0454* (0.0271)	0.0380 (0.0327)	0.0222 (0.0532)	-0.0520 (0.0890)
Female	0.4079** (0.1681)	0.6723*** (0.2299)	0.6539** (0.2671)	0.1311 (0.3452)	-0.5060 (0.5606)
U.S. African/Hispanic/Native	0.4221 (0.5258)	1.0081*** (0.3545)	1.1601*** (0.4426)	0.0120 (0.8411)	-0.8263 (1.1611)
U.S. Asian	-0.5304 (0.4199)	-0.5669 (0.6504)	-0.3260 (1.2512)	0.8638 (0.7201)	1.1711 (0.7276)
International Asian	-0.8445*** (0.2945)	0.1353 (0.4146)	0.9405 (0.7233)	0.4990 (0.6729)	-0.7827 (0.9323)
International Other	0.5034 (0.3172)	0.3781 (0.4131)	1.3523*** (0.5086)	-0.1061 (0.7483)	0.6239 (1.1070)
GRE Verbal Percentile	0.0155*** (0.0060)	0.0281* (0.0160)	0.0753** (0.0344)	-0.0092 (0.0154)	0.0607*** (0.0218)
GRE Quantitative Percentile	0.1547*** (0.0186)	0.0161* (0.0086)	0.0210* (0.0116)	0.0211 (0.0150)	-0.0138 (0.0159)
GRE Analytic Percentile	0.0179*** (0.0042)	0.0199** (0.0094)	0.0030 (0.0134)	0.0175 (0.0127)	-0.0202 (0.0218)
Undergraduate GPA	2.2212*** (0.7219)	0.6968 (0.5476)	1.3648** (0.5513)	1.2545 (0.9703)	1.7772* (0.9487)
TOEFL/IELTS	0.3878 (0.2883)	0.6456 (0.4708)	1.0963** (0.5526)	0.9655* (0.5456)	-0.4656 (0.6112)
Elite U.S. University	0.8850*** (0.2960)	0.3943 (0.3136)	0.6541* (0.3022)	-0.6057 (0.4065)	1.4303** (0.6102)
Elite U.S. Liberal Arts	1.1173*** (0.3813)	0.6276* (0.3754)	1.0690** (0.5390)	-0.6088 (1.4186)	2.3151** (0.6422)
Elite Foreign University	1.0061* (0.6060)	0.4750 (0.8224)	1.7715** (0.7938)	1.6195* (0.8587)	0.3889 (1.9889)
Other Foreign University	0.5442 (0.5765)	0.5777 (0.7686)	1.3732** (0.6902)	0.4188 (0.7157)	-1.0737 (1.3016)
Graduate Degree	0.0467 (0.2238)	0.0733 (0.2583)	0.3128 (0.3280)	0.4049 (0.4806)	0.4892 (0.6399)
Work Experience	0.3932** (0.1880)	0.4027 (0.2993)	0.4972 (0.3295)	0.1495 (0.3778)	1.8201** (0.8564)
#Professor Recommenders	0.0340 (0.1317)	-0.0179 (0.1339)	0.5511** (0.2554)	0.4781* (0.2662)	-0.0981 (0.2660)
#Prolific Recommenders	0.2635*** (0.0854)	0.3025** (0.1262)	0.1880 (0.1576)	0.3284** (0.1544)	0.5901* (0.3145)
#Math Courses	0.1809*** (0.0596)				
Advanced Math	0.2319 (0.1790)				
Missing GRE	0.1899 (0.7646)	-0.3259 (0.3109)	0.6282 (0.4275)	-0.0831 (0.3761)	-1.1514** (0.5702)
Missing Undergraduate GPA	-0.4745 (0.4801)	-0.4917 (0.5440)	-1.5402** (0.6893)	-1.1097* (0.6481)	0.4660 (1.1831)
Missing TOEFL/IELTS	-0.3786 (0.2432)	0.6188 (0.5657)	0.6745 (0.5190)	-0.0440 (0.5099)	-1.7224* (0.9271)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R ²	0.2677	0.1525	0.1862	0.1871	0.2475

H robust standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, #Prolific Recommenders, #Math Courses
- **Economics Negative ($p < .1$):** Covid Year, Winsorized Age, International Asian

A4: Graduate School Logistic Regression AMEs

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0303** (0.0153)	-0.0164 (0.0100)	-0.0431* (0.0261)	-0.0298 (0.0224)	-0.0075 (0.0238)	-0.0218*** (0.0084)
Winsorized Age	-0.0076*** (0.0027)	0.0017* (0.0010)	0.0028 (0.0025)	0.0012 (0.0029)	-0.0022 (0.0038)	0.0016* (0.0009)
Female	0.0312** (0.0126)	0.0254*** (0.0088)	0.0490** (0.0220)	0.0071 (0.0187)	-0.0216 (0.0229)	0.0207*** (0.0075)
U.S. African/Hispanic/Native	0.0323 (0.0402)	0.0381*** (0.0137)	0.0869*** (0.0332)	0.0007 (0.0455)	-0.0352 (0.0487)	0.0384*** (0.0119)
U.S. Asian	-0.0413 (0.0322)	-0.0215 (0.0247)	-0.0244 (0.0936)	0.0468 (0.0391)	0.0499 (0.0313)	0.0062 (0.0185)
International Asian	-0.0649*** (0.0224)	-0.0070 (0.0157)	0.0704 (0.0543)	0.0265 (0.0363)	-0.0334 (0.0393)	0.0069 (0.0132)
International Other	0.0385 (0.0243)	0.0143 (0.0157)	0.1013*** (0.0385)	-0.0057 (0.0406)	0.0266 (0.0471)	0.0238* (0.0132)
GRE Verbal Percentile	0.0012*** (0.0005)	0.0011* (0.0006)	0.0056** (0.0026)	-0.0005 (0.0008)	0.0026*** (0.0010)	0.0015*** (0.0005)
GRE Quantitative Percentile	0.0118*** (0.0014)	0.0006* (0.0003)	0.0016* (0.0009)	0.0011 (0.0008)	-0.0006 (0.0007)	0.0006** (0.0003)
GRE Analytic Percentile	0.0014*** (0.0003)	0.0008** (0.0004)	0.0002 (0.0010)	0.0009 (0.0007)	-0.0009 (0.0009)	0.0007** (0.0003)
Undergraduate GPA	0.1609*** (0.0551)	0.0264 (0.0208)	0.1022** (0.0423)	0.0484 (0.0522)	0.0757* (0.0418)	0.0488*** (0.0176)
TOEFL/IELTS	0.0297 (0.0219)	0.0244 (0.0179)	0.0821** (0.0417)	0.0523* (0.0301)	-0.0198 (0.0259)	0.0326*** (0.0123)
Elite U.S. University	0.0677*** (0.0225)	0.0149 (0.0119)	0.0490* (0.0281)	-0.0328 (0.0264)	0.0630** (0.0271)	0.0233** (0.0090)
Elite U.S. Liberal Arts College	0.0855*** (0.0288)	0.0238* (0.0143)	0.0801** (0.0402)	-0.0330 (0.0761)	0.0987** (0.0480)	0.0429*** (0.0136)
Elite Foreign University	0.0770* (0.0463)	0.0180 (0.0311)	0.1327** (0.0591)	0.0877* (0.0481)	0.0166 (0.0850)	0.0479** (0.0241)
Other Foreign University	0.0416 (0.0441)	0.0219 (0.0201)	0.1026* (0.0527)	0.0227 (0.0391)	0.0458 (0.0556)	0.0273 (0.0221)
Graduate Degree	0.0036 (0.0171)	0.0028 (0.0098)	0.0234 (0.0247)	0.0219 (0.0262)	0.0208 (0.0273)	0.0106 (0.0089)
Work Experience	0.0301** (0.0144)	0.0152 (0.0113)	0.0372 (0.0246)	0.0061 (0.0205)	0.0776** (0.0368)	0.0221** (0.0099)
#Professor Recommenders	0.0026 (0.0101)	-0.0007 (0.0051)	0.0413** (0.0190)	0.0259* (0.0142)	-0.0042 (0.0113)	0.0003 (0.0047)
#Profic Recommenders	0.0202*** (0.0065)	0.0114** (0.0048)	0.0141 (0.0118)	0.0178** (0.0082)	0.0251* (0.0130)	0.0133*** (0.0038)
#Math Courses	0.0129*** (0.0045)					
Advanced Math	0.0177 (0.0137)					
Missing GRE	0.0145 (0.0585)	-0.0123 (0.0118)	0.0470 (0.0319)	-0.0045 (0.0204)	-0.0491* (0.0252)	-0.0062 (0.0088)
Missing Undergraduate GPA	-0.0363 (0.0375)	-0.0186 (0.0206)	-0.1169*** (0.0524)	-0.0601** (0.0355)	0.0200 (0.0505)	-0.0291 (0.0179)
Missing TOEFL/IELTS	-0.0290 (0.0188)	0.0234 (0.0215)	0.0505 (0.0388)	-0.0024 (0.0276)	-0.0734* (0.0417)	0.0108 (0.0135)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic DF P-Value	115.8837	23	0.0000			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, #Prolific Recommenders, #Math Courses
- **Economics Negative ($p < .1$):** Covid Year, Winsorized Age, International Asian

A4: Graduate School Logistic Regression Results (Synthetic)

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.1997 (0.1633)	-0.0323 (0.2215)	0.2028 (0.3078)	-0.4791 (0.3477)	-0.5946 (0.4520)
Winsorized Age	-0.0509** (0.0201)	-0.0198 (0.0174)	0.0576* (0.0307)	-0.0164 (0.0357)	0.1203*** (0.0400)
Female	0.1941 (0.1402)	0.5023 (0.2004)	0.1389 (0.2526)	0.2595 (0.3255)	0.2119 (0.4310)
U.S. African/Hispanic/Native	0.3053 (0.3724)	-0.1457 (0.4303)	-0.2379 (0.4670)	-1.6401 (1.0720)	0.5413 (0.5306)
U.S. Asian	0.5663 (0.3642)	0.6016 (0.4452)	-0.7934 (1.1675)	-0.4305 (0.7939)	0.7122 (0.8863)
International Asian	-0.1428 (0.2169)	0.1630 (0.2803)	0.5780 (0.3740)	-0.4965 (0.4275)	0.1866 (0.6866)
International Other	-0.1231 (0.2311)	0.1395 (0.3046)	0.1097 (0.3237)	-0.1602 (0.3832)	-1.0807 (0.7583)
GRE Verbal Percentile	0.0167*** (0.0046)	0.0247** (0.0117)	0.0164 (0.0127)	-0.0029 (0.0100)	0.0132 (0.0184)
GRE Quantitative Percentile	0.0233*** (0.0090)	-0.0027 (0.0051)	0.0026 (0.0088)	0.0304** (0.0137)	0.0097 (0.0099)
GRE Analytic Percentile	0.0042 (0.0031)	0.0004 (0.0059)	-0.0073 (0.0062)	-0.0042 (0.0101)	0.0069 (0.0103)
Undergraduate GPA	0.2292 (0.4745)	0.6327 (0.4085)	1.3552** (0.5706)	-1.3171** (0.5584)	-0.1802 (0.6152)
TOEFL/IELTS	-0.4673** (0.2157)	0.5141 (0.4185)	-0.7648 (0.6814)	0.2446 (0.7498)	-0.7354 (0.8025)
Elite U.S. University	-0.4850* (0.2560)	-0.2243 (0.3258)	0.0572 (0.3421)	0.4764 (0.4788)	0.3941 (0.4762)
Elite U.S. Liberal Arts College	0.2429 (0.3049)	0.1759 (0.4042)	0.2358 (0.4964)	0.4860 (0.9631)	0.8652 (0.7261)
Elite Foreign University	-0.1622 (0.3015)	-1.1920 (0.7484)	0.7404 (0.5668)	0.4801 (0.7841)	3.0530*** (0.8673)
Other Foreign University	-0.1071 (0.1979)	0.2042 (0.2732)	0.2721 (0.3301)	0.1731 (0.4202)	0.1731 (0.5926)
Graduate Degree	-0.0198 (0.1624)	0.1770 (0.2439)	-0.1601 (0.2765)	-0.8608*** (0.3320)	-0.2769 (0.3814)
Work Experience	0.2525* (0.1441)	-0.0391 (0.2396)	-0.0736 (0.2629)	-0.2650 (0.3417)	-1.1936** (0.4805)
@Professor Recommenders	-0.1059 (0.0909)	0.0011 (0.1138)	0.1828 (0.1473)	0.2181 (0.1817)	-0.4923** (0.1956)
@Prolific Recommenders	0.0039 (0.0956)	0.2189** (0.1108)	-0.1743 (0.1723)	0.0673 (0.1642)	0.5134** (0.2213)
@Math Courses	0.0366 (0.0421)				
Advanced Math	0.1961 (0.1416)				
Missing GRE	-0.1993 (0.4328)	-0.2020 (0.2195)	-0.0011 (0.2689)	-0.0433 (0.3503)	-0.6040 (0.4817)
Missing Undergraduate GPA	0.1531 (0.1548)	0.0895 (0.2291)	-0.2455 (0.3015)	0.6298** (0.3125)	-0.3574 (0.4342)
Missing TOEFL/IELTS	-0.2950* (0.1640)	0.4898 (0.3480)	0.5752 (0.5466)	0.1346 (0.4345)	-2.0781*** (0.7633)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R ²	0.0524	0.0396	0.0753	0.1046	0.2230

Robust standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive** ($p < .1$): GRE Verbal Percentile, GRE Quantitative Percentile, Work Experience
- **Economics Negative** ($p < .1$): Winsorized Age, TOEFL/IELTS, Elite U.S. University, Missing TOEFL/IELTS

A4: Graduate School Logistic Regression AMEs (Synthetic)

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0204 (0.0167)	-0.0014 (0.0099)	0.0192 (0.0291)	-0.0298 (0.0216)	-0.0345 (0.0262)	-0.0089 (0.0086)
Winsorized Age	-0.0052** (0.0021)	-0.0009 (0.0008)	0.0054* (0.0029)	-0.0010 (0.0022)	0.0070*** (0.0023)	0.0004 (0.0008)
Female	0.0108 (0.0143)	0.0041 (0.0239)	0.0131 (0.0294)	0.0162 (0.0251)	0.0123 (0.0078)	0.0099 (0.0078)
U.S. African/Hispanic/Native	0.0312 (0.0381)	-0.0065 (0.0192)	-0.0225 (0.0441)	-0.1021 (0.0673)	0.0314 (0.0306)	-0.0137 (0.0147)
U.S. Asian	0.0579 (0.0372)	0.0268 (0.0199)	-0.0750 (0.1101)	-0.0268 (0.0495)	0.0413 (0.0516)	0.0241 (0.0179)
International Asian	-0.0146 (0.0222)	0.0073 (0.0125)	0.0546 (0.0353)	-0.0309 (0.0263)	0.0108 (0.0399)	0.0075 (0.0112)
International Other	-0.0126 (0.0236)	0.0062 (0.0136)	0.0104 (0.0306)	-0.0100 (0.0237)	-0.0627 (0.0443)	-0.0014 (0.0106)
GRE Verbal Percentile	0.0017*** (0.0005)	0.0011** (0.0005)	0.0015 (0.0012)	-0.0002 (0.0006)	0.0008 (0.0010)	0.0008** (0.0004)
GRE Quantitative Percentile	0.0024*** (0.0009)	-0.0001 (0.0002)	0.0003 (0.0008)	0.0019*** (0.0009)	0.0006 (0.0006)	0.0003 (0.0002)
GRE Analytic Percentile	0.0004 (0.0003)	0.0000 (0.0003)	-0.0007 (0.0009)	-0.0003 (0.0006)	0.0004 (0.0006)	0.0000 (0.0002)
Undergraduate GPA	0.0234 (0.0485)	0.0282 (0.0223)	0.1281** (0.0553)	-0.0020** (0.0358)	-0.0058 (0.0357)	0.0130 (0.0158)
TOEFL/IELTS	-0.0478** (0.0221)	0.0229 (0.0187)	-0.0723 (0.0645)	0.0152 (0.0466)	-0.0427 (0.0469)	0.0050 (0.0187)
Elite U.S. University	-0.0496* (0.0261)	-0.0100 (0.0145)	0.0054 (0.0244)	0.0297 (0.0297)	0.0229 (0.0274)	0.0036 (0.0108)
Elite U.S. Liberal Arts College	0.0248 (0.0312)	0.0078 (0.0180)	0.0223 (0.0469)	0.0303 (0.0602)	0.0502 (0.0418)	0.0217 (0.0160)
Elite Foreign University	-0.0166 (0.0308)	-0.0532 (0.0337)	0.0700 (0.0537)	0.0299 (0.0490)	0.1771*** (0.0484)	0.0158 (0.0179)
Other Foreign University	-0.0111 (0.0202)	0.0091 (0.0122)	0.0018 (0.0321)	0.0015 (0.0267)	0.0175 (0.0344)	0.0116 (0.0103)
Graduate Degree	-0.0020 (0.0166)	0.0079 (0.0109)	-0.0151 (0.0261)	-0.0542*** (0.0206)	-0.0161 (0.0220)	-0.0093 (0.0085)
Work Experience	0.0258* (0.0148)	-0.0017 (0.0107)	-0.0070 (0.0248)	-0.0165 (0.0131)	-0.0692** (0.0275)	-0.0093 (0.0086)
#Professor Recommenders	-0.0108 (0.0093)	0.0041 (0.0051)	-0.0173 (0.0140)	0.0136 (0.0114)	-0.0286*** (0.0110)	-0.0027 (0.0042)
#Profic Recommenders	0.0004 (0.0067)	0.0008* (0.0050)	-0.0165 (0.0164)	0.0042 (0.0102)	0.0298** (0.0128)	0.0083* (0.0045)
#Math Courses	0.0237 (0.0043)					
Advanced Math	0.0201 (0.0145)					
Missing GRE	-0.0204 (0.0443)	-0.0090 (0.0088)	-0.0001 (0.0254)	-0.0027 (0.0218)	-0.0390 (0.0239)	-0.0098 (0.0079)
Missing Undergraduate GPA	0.0157 (0.0158)	0.0040 (0.0102)	0.0232 (0.0285)	0.0392* (0.0201)	-0.0207 (0.0254)	0.0019 (0.0083)
Missing TOEFL/IELTS	-0.0302* (0.0168)	0.0218 (0.0156)	0.0544 (0.0517)	0.0084 (0.0272)	-0.1205*** (0.0434)	0.0063 (0.0130)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic DF P-Value	37.4658	23	0.0290			

Robust standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive** ($p < .1$): GRE Verbal Percentile, GRE Quantitative Percentile, Work Experience
- **Economics Negative** ($p < .1$): Winsorized Age, TOEFL/IELTS, Elite U.S. University, Missing TOEFL/IELTS

A4: Graduate School Lasso Logistic Regression Results

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3330	-0.3123	-0.3498	-0.2378	
Winsorized Age	-0.0788	0.0266	0.0200		
Female	0.3352	0.5096	0.5068		-0.0654
U.S. African/Hispanic/Native	0.2638	0.8041	0.7157		
U.S. Asian	-0.2802	-0.2918		0.0413	0.1155
International Asian	-0.6744	-0.0726		0.0465	
International Other	0.4729	0.1946	0.7240	-0.1087	
GRE Verbal Percentile	0.0139	0.0252	0.0272		0.0343
GRE Quantitative Percentile	0.1349	0.0119	0.0187	0.0154	
GRE Analytic Percentile	0.0166	0.0208		0.0080	
Undergraduate GPA	1.9977	0.4621	0.7414	0.1714	
TOEFL/IELTS	0.4009	0.3295	0.5199	0.5151	
Elite U.S. University	0.6886	0.2709	0.3267		0.4606
Elite U.S. Liberal Arts College	0.8956	0.5199	0.5985		0.4470
Elite Foreign University	0.3724		0.0414	0.4343	
Other Foreign University					
Graduate Degree			0.0978		
Work Experience	0.2835	0.3447	0.3273		0.5559
#Professor Recommenders			0.3739	0.2322	
#Prolific Recommenders	0.2243	0.2362	0.1651	0.2213	0.0632
#Math Courses	0.1493				
Advanced Math	0.2272				
Missing GRE		-0.2304		-0.0527	-0.5416
Missing Undergraduate GPA					
Missing TOEFL/IELTS	-0.2364	0.2852	0.0796		
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R ²	0.2642	0.1456	0.1589	0.1486	0.1362
CV Pseudo-R ²	0.2297	0.0912	0.0592	0.0850	0.0537

- **Economics Positive:** Female, U.S. African/Hispanic/Native, International Other, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, TOEFL/IELTS, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, # Prolific Recommenders, #Math Courses, Advanced Math
- **Economics Negative:** Covid Year, Winsorized Age, U.S. Asian, International Asian, Missing TOEFL/IELTS

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